Bangladesh University of Engineering and Technology
Department of Computer Science & Engineering

Digital Logic Design Question Bank

CSE 205 · Digital Logic Design

Year Coverage: 2011-12 to 2024–25
Topics: Boolean Algebra, Logic Gates, Minimization, Combinational Logic, Flip-Flops, Sequential Circuits, Registers & Counters, Memory & PLDs

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Digital Logic Design

Boolean Algebra, Theorems & Logic Gates

Postulates, Duality, Truth Tables, Gate Implementations

S1 Boolean Algebra, Theorems & Logic Gates

Boolean Algebra & Simplification

Q1. Using laws and theorems of Boolean algebra, prove and simplify the following expression:

$$xy + x'z + yz = xy + x'z$$

Apply relevant laws and theorems to show that the term yz is redundant and obtain the simplified form. **(7 marks)** [CSE 205 | L-2/T-1 | 2024–25]

Q2. Analyze whether the dual and the complement of the following function are the same:

$$F = ABC + A(B + C) + A'B'C'$$

Justify your answer. **(8 marks)** [CSE 205 | L-2/T-1 | 2023–24]

Q3. Using Boolean algebra, show that $F = xyz' + wx'y' + (x' + z + w)(x' + z + w') + xyz + wx'y$ can be simplified to $F = xy + wx' + x' + z$. **(11 marks)** [CSE 205 | L-2/T-1 | 2023–24]

Q4. Simplify the following Boolean expressions to a minimum number of literals: $(A + B)'(A' + B')'$. **(10 marks)** [CSE 205 | L-2/T-1 | 2021–22]

Q5. Simplify the following expression to POS and SOP: $ACD' + C'D + AB' + ABCD$. **(5 marks)** [CSE 205 | L-2/T-1 | 2018–19]

Q6. Simplify the following expression to POS and SOP: $BCD' + ABC' + ACD$. **(8 marks)** [CSE 205 | L-2/T-1 | 2017–18]

Q7. Prove the following Boolean algebra theorem:

$$(x + y)(x' + z)(y + z) = (x + y)(x' + z)$$

(10 marks) [CSE 205 | L-2/T-1 | 2015–16]

Q8. How many distinct Boolean functions are there of n Boolean variables? **(5 marks)** [CSE 205 | L-2/T-1 | 2015–16]

Q9. Simplify the Boolean function using algebraic manipulation:

$$f = A'BC + AB'C + ABC' + ABC$$

(10 marks) [CSE 205 | L-2/T-1 | 2015–16]

Q10. Simplify the following Boolean function using algebraic manipulation:

$$(A + BC)' + AB'$$

(7 marks) [CSE 205 | L-2/T-1 | 2015–16]

Q11. State and prove the consensus theorem using Boolean algebra. **(8 marks)** [CSE 205 | L-2/T-1 | 2014–15]

Q12. Simplify the following function using Boolean algebra:

$$F(A, B, C) = A'BC + AB'C + ABC$$

(7 marks) [CSE 205 | L-2/T-1 | 2014–15]

Q13. Simplify the following Boolean expressions to a minimum number of literals using Boolean algebra:

(a) $AB + A(CD + CD') + A'$

(b) $(A + B)'(A' + B')'$

(10 marks) [CSE 205 | L-2/T-1 | 2013–14]

Q14. The Boolean expression for an X-OR gate is $AB' + A'B$. With this as a starting point, use DeMorgan's theorems and other rules or laws to find a simplified expression for an X-NOR gate. **(8 marks)** [CSE 205 | L-2/T-1 | 2011–12]

Q15. Give an example of the non-associativity of the 3-input NOR operator. **(8 marks)** [CSE 205 | L-2/T-1 | 2011–12]

Q16. Prove that $x(x + y) = x$. **(5 marks)** [CSE 205 | L-2/T-1 | 2011–12]

De Morgan's Theorem

Q1. Demonstrate the validity of the following identities by means of truth tables:

(a) De Morgan's theorem for three variables: $(x + y + z)' = x'y'z'$

(b) The distributive law: $x(y + z) = xy + xz$

(10 marks) [CSE 205 | L-2/T-1 | 2019–20]

Q2. Given De Morgan's theorem for two variables: $(A + B)' = A'B'$, prove De Morgan's theorem for three variables by successive substitution. **(10 marks)** [CSE 205 | L-2/T-1 | 2013–14]

Q3. State and prove De Morgan's theorem using a truth table. **(10 marks)** [CSE 205 | L-2/T-1 | 2012–13]

Logic Gates, Universal Gates & Definitions

Q1. Why are NAND and NOR gates called universal gates? Explain how hardware cost is related to Boolean simplification. If a function is already minimal, how can you verify it? **(10 marks)** [CSE 205 | L-2/T-1 | 2024–25]

Q2. How does a three-state gate work? What can be the advantage of using a three-state gate? **(6 marks)** [CSE 205 | L-2/T-1 | 2024–25]

Q3. What do you mean by Universal gate? Show that both NAND gate and NOR gate are universal gates. **(14 marks)** [CSE 205 | L-2/T-1 | 2022–23, 2013–14]

↪ *Similar: 2019–20*

Q4. Show the circuit diagram of the expression $AB + CDE$ using only NAND gates. **(10 marks)** [CSE 205 | L-2/T-1 | 2021–22]

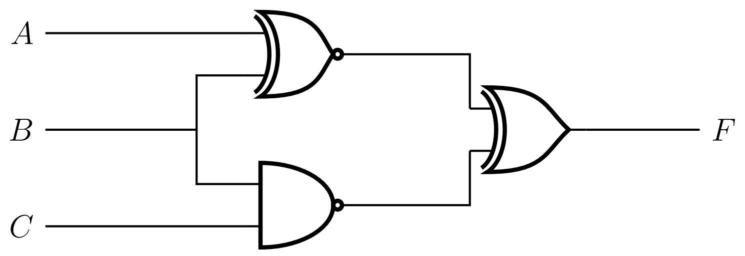
Q5. A majority circuit is a combinational circuit whose output is equal to 1 if the input variables have more 1's than 0's, and 0 otherwise. Design a 3-input majority circuit by finding the circuit's truth table, simplified Boolean equation, and a logic diagram. Use only NAND gate(s) for your design. **(18 marks)** [CSE 205 | L-2/T-1 | 2019–20]

Q6. Show that NAND gate is a universal logic gate. Also show that NAND operation does not follow the associative law. **(5+5 marks)** [CSE 205 | L-2/T-1 | 2015–16]

Q7. Define: Odd Function, Even Function, Don't-care conditions. **(5 marks)** [CSE 205 | L-2/T-1 | 2013–14]

Circuit Analysis

Q1. Analyze the following circuit and determine the output F as sum of minterms and product of maxterms.



(9 marks) [CSE 205 | L-2/T-1 | 2011–12]

Two-Level & Multi-Level Logic Implementation

Q1. Simplify the following function and draw the circuit using only NAND gates:

$$F(w, x, y, z) = \Sigma(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$$

(16 marks) [CSE 205 | L-2/T-1 | 2023–24]

Q2. Implement $F(x, y, z) = x'y'z' + xyz'$ using two-level forms of logic by using NAND-AND gates. Assume that the complement of x, y and z are readily available. **(14 marks)** [CSE 205 | L-2/T-1 | 2023–24]

Q3. Express $f(A, B, C, D) = \Sigma(0, 4, 8, 9, 10, 11, 12, 14)$ with the two-level forms of logic "NAND-AND" and "OR-NAND". **(6 marks)** [CSE 205 | L-2/T-1 | 2018–19]

Q4. Implement the function $F(w,x,y,z) = w'x' + w'x'z + w'yz'$ using two-level forms of logic by

(a) NOR gates only

(b) AND-NOR

(c) NAND gates only

(d) OR-NAND

(20 marks) [CSE 205 | L-2/T-1 | 2014–15]

Q5. Show the truth table for the following function:

$$F(x,y,z) = (x+y+z)(x+y+z')(x+y'+z)(x'+y+z)$$

(8 marks) [CSE 205 | L-2/T-1 | 2014–15]

Q6. Given function $F = A'B + BC + AC$:

(a) Implement F using AND, OR and NOT gates.

(b) Minimize F in SOP form using K-Map.

(c) Implement the minimized F using only NAND gates.

(15 marks) [CSE 205 | L-2/T-1 | 2013–14]

Q7. Given the Boolean function $F = xy + x'y' + y'z$, implement it with (i) AND, OR and NOT gates, and (ii) only NAND gates. **(5+5 marks)** [CSE 205 | L-2/T-1 | 2012–13]

Q8. Using the two-level forms of logic (i) NOR-OR, and (ii) OR-NAND, implement the following Boolean function F:

$$F(A,B,C,D) = \Sigma\,(0,4,8,9,10,11,12,14)$$

Assume that both the normal and complement inputs are available. **(6+6 marks)** [CSE 205 | L-2/T-1 | 2011–12]

NOR Gate Implementation

Q1. Express $f(A,B,C) = \Pi\,(0,2,3,5,7)$ with NOR gates. **(5 marks)** [CSE 205 | L-2/T-1 | 2020–21]

Q2. Express $f(A,B,C,D) = \text{SOP}(0,4,8,9,10,11,12,14)$ with the two-level forms of logic "NOR-OR" and "AND-NOR". **(8 marks)** [CSE 205 | L-2/T-1 | 2017–18]

Q3. Using NOR gates, evaluate the following expression:

$$p \oplus (q \vee r) \oplus ((p \wedge q) \rightarrow r)$$

(5 marks) [CSE 205 | L-2/T-1 | 2016–17]

Q4. Implement the following Boolean function using only NOR gates:

$$f = xy + yz$$

Your implementation should use as few NOR gates as possible. **(6 marks)** [CSE 205 | L-2/T-1 | 2015–16]

Q5. Implement the following Boolean function together with the don't-care conditions using only two-input NOR gates (complements of literals are available as input):

$$F(A,B,C,D) = \Sigma\,(0,1,9,11), \quad d(A,B,C,D) = \Sigma\,(2,8,10,14,15)$$

(15 marks) [CSE 205 | L-2/T-1 | 2013–14]

XOR Properties

Q1. Show that the dual of the exclusive-OR is equal to its complement. **(7 marks)** [CSE 205 | L-2/T-1 | 2022–23, 2019–20]

Q2. Show that XOR operation is not distributive over AND operation. **(7 marks)** [CSE 205 | L-2/T-1 | 2015–16]

Q3. Why is the XOR operation called an odd function? Convert the following Boolean expression to a canonical maxterm form:

$$a \oplus b \oplus c \oplus d$$

(10 marks) [CSE 205 | L-2/T-1 | 2015–16]

Parity Checker and Generator

Q1. In a network, for a 3-bit data transmission, an error will be generated if there is an odd parity. Design this logical circuit. **(5 marks)** **[CSE 205 | L-2/T-1 | 2020–21]**

Q2. What are the differences between an odd parity checker and an odd parity generator? **(8 marks)** **[CSE 205 | L-2/T-1 | 2018–19]**

Q3. For a 3-bit Excess-3 code, design an odd parity checker and odd parity generator. **(8 marks)** **[CSE 205 | L-2/T-1 | 2017–18]**

Q4. Design a negative logic 4-bit parity checker and generator. **(6 marks)** **[CSE 205 | L-2/T-1 | 2016–17]**

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Digital Logic Design

Canonical & Standard Forms

Minterms, Maxterms, SOP, POS, Conversion

S2 Canonical Forms & Boolean Functions

Canonical vs Standard Forms

Q1. Compare the different types of canonical and standard forms used in Boolean Algebra with examples. **(8 marks)** **[CSE 205 | L-2/T-1 | 2023–24]**

Q2. (a) What are the differences between canonical form and standard form?

(b) Which form is preferable when implementing a Boolean function with gates? Why?

(c) Which form is obtained when reading a function from a truth table?

(8+4+3 marks) **[CSE 205 | L-2/T-1 | 2012–13]**

Minterms, Maxterms

Q1. Express the following function in terms of minterms:

$$F = (x' + y)(x + z)(y + z)$$

(10 marks) **[CSE 205 | L-2/T-1 | 2023–24]**

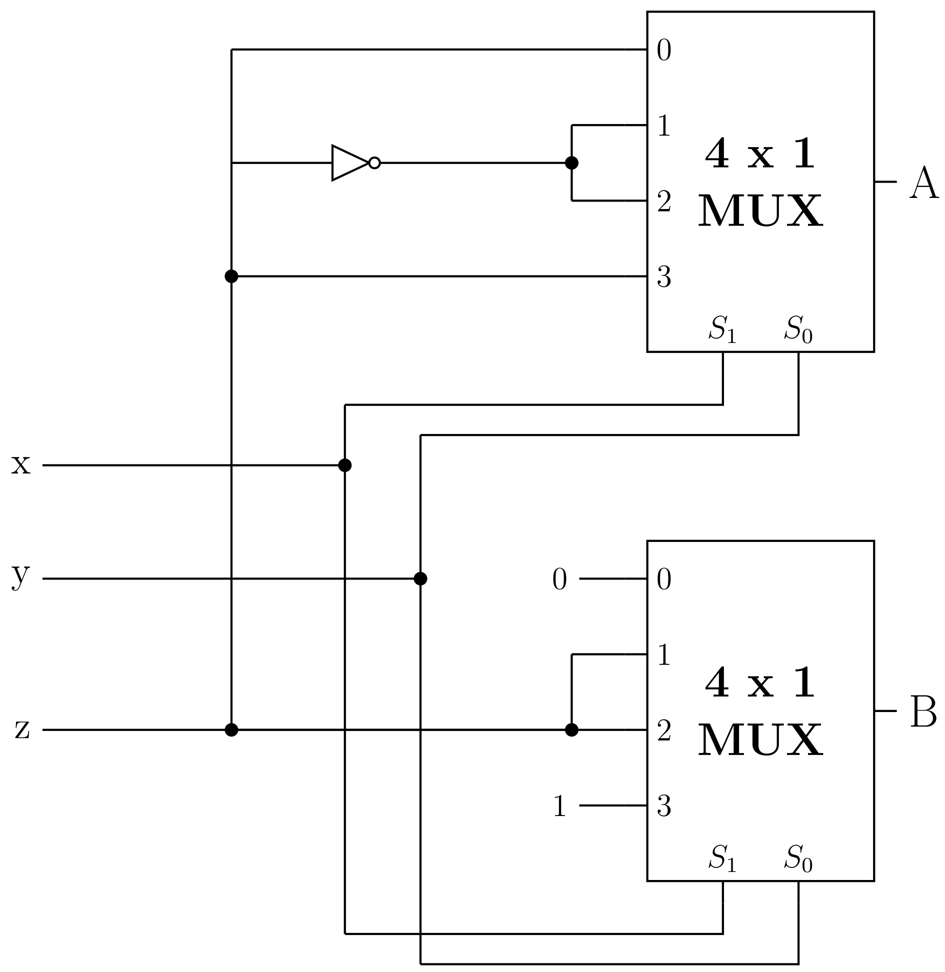
Q2. Convert the Boolean function $F = xy + x'z$ as a product of maxterms using the distributive law. **(10 marks)** **[CSE 205 | L-2/T-1 | 2022–23]**

Q3. Using Boolean Algebra, express the following Boolean function as a sum of minterms:

$$F(A, B, C) = A + B'C$$

(7 marks) **[CSE 205 | L-2/T-1 | 2021–22]**

Q4. Determine the output functions A as the sum of minterms and the output function B as the product of maxterms:



(16 marks) **[CSE 205 | L-2/T-1 | 2014–15]**

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Digital Logic Design

Minimization Techniques

K-Maps (2/3/4/5-variable), Quine-McCluskey, Don't-Care Conditions

S3 Minimization Techniques

Prime Implicants

Q1. Differentiate between prime implicant and essential prime implicant of a K-Map. **(7 marks)** [CSE 205 | L-2/T-1 | 2023–24, 2011–12]

Q2. Find all the prime implicants for the following Boolean function and determine which are essential:

$$F(A, B, C, D) = \Sigma\,(0, 2, 3, 5, 7, 8, 9, 10, 11, 13, 15)$$

(11 marks) [CSE 205 | L-2/T-1 | 2014–15]

Karnaugh Map

Q1. Simplify the following Boolean function into (a) sum-of-products form and (b) product-of-sums form:

$$F(A, B, C, D) = \Sigma\,(0, 1, 2, 5, 8, 9, 10)$$

(14 marks) [CSE 205 | L-2/T-1 | 2022–23]

Q2. Using K-Map, minimize $f(v, w, x, y, z) = \Sigma\,(2, 7, 8, 15, 22, 23, 25, 29) + d(6, 9, 13, 18, 24, 31)$. Find the Prime Implicants and Essential Prime Implicants, and the Minimal Sums. Write down the responsible minterm for each Essential Prime Implicant. **(15 marks)** [CSE 205 | L-2/T-1 | 2020–21]

Q3. Convert the following Boolean function from sum-of-products form to a simplified product-of-sums form, then draw a logic circuit for the POS form:

$$F(w, x, y, z) = \Sigma\,(0, 1, 2, 5, 7, 11, 13)$$

(20 marks) [CSE 205 | L-2/T-1 | 2019–20]

Q4. Using K-Map, minimize $f(w, x, y, z) = \Sigma\,(1, 2, 3, 7, 10, 12, 14) + d(0, 6, 8, 13)$. Find the Prime Implicants and Essential Prime Implicants. Write down the responsible minterm for each Essential Prime Implicant. **(12 marks)** [CSE 205 | L-2/T-1 | 2018–19]

Q5. Using K-Map, minimize $f(w, x, y, z) = \text{SOP}(1, 2, 4, 7, 9, 10, 12, 14) + d(0, 6, 8, 13)$. Find the Prime Implicants and Essential Prime Implicants. Write down the responsible minterm for each Essential Prime Implicant. **(12 marks)** [CSE 205 | L-2/T-1 | 2017–18]

Q6. Using K-map, solve $f(x_1, x_2, x_3, x_4, x_5) = \Sigma\,(0, 2, 4, 5, 6, 7, 8, 10, 14, 17, 18, 21, 29, 31) + d\Sigma\,(11, 20, 22)$. Find the essential prime implicants. (Show the reduction steps.) **(15 marks)** [CSE 205 | L-2/T-1 | 2016–17]

Q7. Using K-maps, find the simplified product of sums form of the following Boolean function:

$$f(w, x, y, z) = \Sigma\,(1, 5, 8, 12, 14, 15), \quad \text{don't-care minterms: } d = \Sigma\,(3, 11)$$

(10 marks) [CSE 205 | L-2/T-1 | 2015–16]

Q8. Using K-maps, find the simplified sum of products form of the following Boolean function:

$$f(a, b, c, d, e, f) = \Sigma\,(1, 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31)$$

Don't-care minterms: $d = \Sigma\,(6, 20, 24, 25, 26, 27)$. **(12 marks)** [CSE 205 | L-2/T-1 | 2015–16]

Q9. Simplify the following Boolean functions using K-map:

(a) $F(A, B, C, D) = AB'C + A'B'D' + ABD + ACD' + AB'C' + A'BC'D$

(b) $F(A, B, C, D) = \Sigma\,(0, 2, 5, 7, 8, 15)$ with don't-care conditions $d(A, B, C, D) = \Sigma\,(10, 14)$

(15 marks) [CSE 205 | L-2/T-1 | 2013–14]

Q10. Minimize the following function in both SOP and POS forms using Karnaugh map:

$$f(A, B, C, D) = \Sigma\, m(0, 4, 7, 9, 13) + d(5, 8, 15)$$

(12 marks) [CSE 205 | L-2/T-1 | 2012–13]

Tabulation Method (Quine-McCluskey)

Q1. Simplify the following Boolean function to product-of-sums form:

$$F = \Sigma\,(3, 4, 6, 7, 9, 11, 12, 14, 15)$$

What is the main purpose of the tabulation method (Quine-McCluskey method) in digital logic design, and how does it systematically help in minimizing Boolean expressions compared to the Karnaugh map method? **(15 marks)** **[CSE 205 | L-2/T-1 | 2024–25]**

Q2. Using tabular method, find the prime implicants and essential prime implicants of the following Boolean function:

$$F(A, B, C, D) = \Sigma\,(0, 2, 3, 5, 7, 8, 9, 10, 11, 13, 15)$$

(15 marks) **[CSE 205 | L-2/T-1 | 2022–23]**

Q3. Determine the prime implicants of the following function using the tabulation (tabular) method:

$$F(w, x, y, z) = \Sigma\,(0, 1, 2, 8, 10, 11, 14, 15)$$

(15 marks) **[CSE 205 | L-2/T-1 | 2021–22]**

Q4. Compare the Tabulation method and the Karnaugh map for minimizing Boolean functions based on their merits and demerits. **(15 marks)** **[CSE 205 | L-2/T-1 | 2019–20]**

Q5. Using the Tabulation method, simplify the following Boolean function by first finding the essential prime implicants:

$$F(A, B, C, D) = \Sigma\,(1, 3, 4, 5, 10, 11, 12, 13, 14)$$

(28 marks) **[CSE 205 | L-2/T-1 | 2019–20]**

Q6. Minimize the following function using the Quine-McCluskey (Q-M) method. Find essential prime implicants and all prime implicants:

$$f(A, B, C, D, E) = \Sigma\,(0, 1, 2, 8, 9, 15, 17, 21, 24, 25, 27, 31)$$

(20 marks) **[CSE 205 | L-2/T-1 | 2016–17]**

Q7. When is the tabulation method more preferable than the K-Map method for simplifying a Boolean function? **(5 marks)** **[CSE 205 | L-2/T-1 | 2014–15]**

Q8. Simplify the following Boolean function by using the tabulation method:

$$F(A, B, C, D) = \Sigma\,(0, 1, 2, 8, 10, 11, 14, 15)$$

(15 marks) **[CSE 205 | L-2/T-1 | 2011–12]**

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Digital Logic Design

Combinational Logic

Adders, Subtractors, Comparators, Decoders, Encoders, MUX, DEMUX

S4 Combinational Logic

Number System & Arithmetic Operations

Q1. Describe the steps to find $M - N$ using $(r-1)$'s complement, where M and N are two n-digit unsigned numbers and r is the base. **(13 marks)** [CSE 205 | L-2/T-1 | 2022–23, 2021–22]

Q2. Explain the difference between binary and gray encoding and the benefits of each. **(9 marks)** [CSE 205 | L-2/T-1 | 2022–23]

Q3. Write the rules of subtraction for two n-digit numbers M and N in base r. Consider two cases: $M \geq N$ and $M < N$. Find the value of $9 - 6.9 - 9.8$ using the complement rule where the base is Hexadecimal. **(2+5 marks)** [CSE 205 | L-2/T-1 | 2017–18]

Q4. What is the difference between r's complement and $(r-1)$'s complement (where r is the base)? **(2 marks)** [CSE 205 | L-2/T-1 | 2016–17]

Q5. Convert $(0.625)_{10}$ to binary. Show every step of your calculation. **(4 marks)** [CSE 205 | L-2/T-1 | 2014–15]

Q6. A decimal number N is represented in a weighted binary code as `0011 0100 1010 1100`. Write the decimal digits of N, if the following weights are used for the weighted binary code: 6, 3, 1, 1. **(4 marks)** [CSE 205 | L-2/T-1 | 2014–15]

Q7. You are given two numbers: $(305.E)_{16}$ and $(109.6875)_{10}$.

(a) Convert the given numbers into binary.

(b) Subtract the latter number from the former using binary arithmetic.

(c) Convert the result of the subtraction into octal.

(15 marks) [CSE 205 | L-2/T-1 | 2013–14]

Q8. Perform the following binary subtraction: $11000011 - 10101010$. **(6 marks)** [CSE 205 | L-2/T-1 | 2012–13]

Q9. Using 2's complement, subtract decimal -70 from decimal -50. **(9 marks)** [CSE 205 | L-2/T-1 | 2012–13]

Q10. Convert $(41.6875)_{10}$ to binary. **(5 marks)** [CSE 205 | L-2/T-1 | 2011–12]

Q11. Perform the subtraction with the following binary unsigned numbers using (i) 2's complement, (ii) 1's complement: $11010 - 1101$. **(10 marks)** [CSE 205 | L-2/T-1 | 2011–12]

Code Converter

Q1. A consensus circuit is a combinational circuit whose output is equal to 1 if exactly two inputs are 1, and 0 otherwise. Design a 3-input consensus circuit by finding the circuit's truth table, simplified Boolean equation and a circuit diagram. **(12 marks)** [CSE 205 | L-2/T-1 | 2023–24]

Q2. Design a combinational circuit that converts a three-bit gray code to a three-bit binary number. Use only a minimum number of gates in your design. **(20 marks)** [CSE 205 | L-2/T-1 | 2022–23]

Q3. Design a combinational circuit that converts a decimal digit represented with "Random Code" to "BCD Code". Use the K-Map method for logic simplification and show the circuit diagram.

Decimal Digit	Random Code ABCD	BCD Code wxyz
0	0000	0000
1	0011	0001
2	0101	0010
3	0110	0011
4	1010	0100
5	1011	0101
6	1100	0110
7	1101	0111
8	1110	1000
9	1111	1001

(20 marks) [CSE 205 | L-2/T-1 | 2021–22]

Binary Adder & Subtractor

Q1. Design a 4-bit binary subtractor circuit using full adders.

(a) Construct and draw the logic diagram of the subtractor circuit.

(b) Explain how subtraction is performed using 2's complement representation.(Understand level)

(15 marks) [CSE 205 | L-2/T-1 | 2024–25]

Q2. Design a full subtractor circuit with three inputs x, y, B_{in} and two outputs Diff and B_{out}. The circuit subtracts $x - y - B_{in}$, where B_{in} is the input borrow and B_{out} is the output borrow. **(18 marks)** [CSE 205 | L-2/T-1 | 2019–20]

Q3. Using 4-bit binary adder and XOR gate, design a circuit which takes two 4-bit BCD inputs A and B, and returns $A+1$ if the control input is 00, A' if the control input is 01, $A'+1$ when the control input is 10, and A when the control input is 11.**(13 marks)** [CSE 205 | L-2/T-1 | 2018–19]

Q4. Write the advantages and disadvantages of serial and parallel adders. **(4 marks)** [CSE 205 | L-2/T-1 | 2015–16]

Q5. Design a four-bit binary adder using a 1-bit full-adder circuit. How can it be modified to use as a four-bit subtractor? **(15 marks)** [CSE 205 | L-2/T-1 | 2013–14]

Q6. Starting with a truth table, design a full subtractor. **(20 marks)** [CSE 205 | L-2/T-1 | 2012–13]

BCD Adder-Subtractor

Q1. Design a BCD adder-subtractor which takes two 4-bit BCD numbers A and B. Addition and subtraction is selected by a selector bit S: when $S = 0$ the circuit produces $A + B$; when $S = 1$ it produces $A - B$. Explain the operations $9 + 7$ and $9 - 7$ with your designed circuit. **(12 marks)** [CSE 205 | L-2/T-1 | 2020–21, 2017–18, 2016–17]

Q2. Design a 4-bit BCD adder using 4-bit binary adders and basic logic gates. Use a block diagram to draw a binary adder. **(15 marks)** [CSE 205 | L-2/T-1 | 2014–15]

Carry Look-Ahead Adder (CLA)

Q1. Explain the working principle of a Carry Look-Ahead (CLA) adder. Derive the expressions for carry generation (C_1, C_2, C_3, C_4) using generate and propagate functions. **(12 marks)** [CSE 205 | L-2/T-1 | 2024–25]

Q2. What is the benefit of using an adder with carry lookahead logic? Derive expressions of different carry of a 4-bit adder with carry lookahead logic. **(15 marks)** [CSE 205 | L-2/T-1 | 2023–24]

Q3. Design a 4 bit Carry Look Ahead (CLA) adder which will take $X_3X_2X_1X_0$ and $Y_3Y_2Y_1Y_0$ as input, and produce their sum. Show the steps and the role of different *carries* of this CLA. **(13 marks)** [CSE 205 | L-2/T-1 | 2020–21]

Q4. Design a 3-bit CLA circuit and show all types of "carry" for the addition $3 + 4$ with your designed CLA. **(8+4 marks)** [CSE 205 | L-2/T-1 | 2017–18]

Q5. For a carry look-ahead adder circuit, what advantage do we get if we use it instead of a 4-bit simple full adder? For a NOT gate with 2 ns propagation delay and other basic gates with 4 ns propagation delay, what will be the total propagation delay? **(5 marks)** [CSE 205 | L-2/T-1 | 2016–17]

Q6. What is the problem with the binary parallel adder circuit? Design and implement a 4-bit carry look-ahead adder circuit. If the propagation delay is 1 ns for a NOT gate and 2 ns for other basic gates, what is the propagation delay of the CLA adder to produce its final output? **(18 marks)** [CSE 205 | L-2/T-1 | 2015–16]

Q7. What are the advantages of a CLA over a normal binary four-bit adder? Assume that the exclusive-OR gate has a propagation delay of 10 ns and that the AND or OR gates have a propagation delay of 5 ns. What is the total propagation delay time in the four-bit CLA adder? **(10 marks)** [CSE 205 | L-2/T-1 | 2013–14]

2's Complement Circuit

Q1. Using only half adders, design a circuit that outputs the 1's complement of a 4-bit input when the control input C is 1, and passes the input unchanged when C is 0. **(6 marks)** [CSE 205 | L-2/T-1 | 2023–24]

Q2. Design a circuit that takes a 4-bit number as input and produces the 2's complement of the given number. **(20 marks)** [CSE 205 | L-2/T-1 | 2011–12]

Overflow Detection

Q1. What is overflow in binary arithmetic? Explain how overflow is detected in signed addition and signed subtraction. Provide suitable examples. **(11 marks)** [CSE 205 | L-2/T-1 | 2024–25]

Q2. How do you check for overflow in the case of addition of two binary numbers in both signed and unsigned representations? **(5 marks)** [CSE 205 | L-2/T-1 | 2013–14]

Binary Multiplier

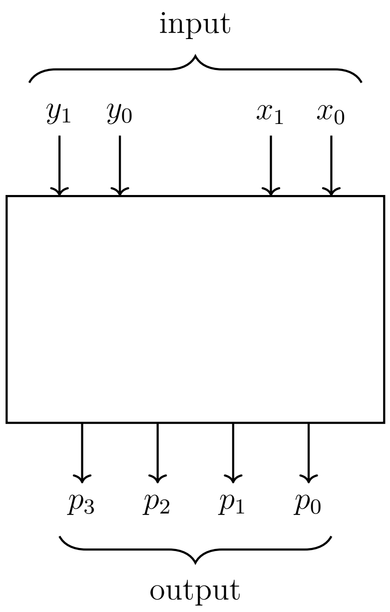
Q1. Discuss and design a 2-bit by 2-bit binary multiplier. **(15 marks)** [CSE 205 | L-2/T-1 | 2021–22]

Q2. Design a circuit with a 4-bit binary adder which returns $A \times B$ where A and B are 3-bit binary numbers. **(10 marks)** [CSE 205 | L-2/T-1 | 2018–19]

Q3. Using 4-bit adders, design a 3-bit multiplier for 3-bit variables $X(X_1, X_2, X_3)$ and $Y(Y_1, Y_2, Y_3)$. Explain the design. **(15 marks)** [CSE 205 | L-2/T-1 | 2016–17]

Q4. How would you implement a one-bit by one-bit binary multiplier using basic gates? **(5 marks)** [CSE 205 | L-2/T-1 | 2013–14]

Q5. The figure below represents a multiplier that takes two 2-bit binary numbers x_1x_0 and y_1y_0 as inputs, and produces an output binary number $p_3p_2p_1p_0$ equal to the arithmetic product of the two input numbers. Design the logic circuit for the multiplier. **(20 marks)** [CSE 205 | L-2/T-1 | 2012–13]



Incrementer

Q1. Design a 4-bit combinational circuit incrementer (a circuit that adds 1 to a 4-bit binary number) using only half adders. **(10 marks)** [CSE 205 | L-2/T-1 | 2021–22]

Q2. Design a 4-bit incrementer circuit using 4 half-adders. Note that the incrementer circuit adds 1 to a binary number. **(10 marks)** [CSE 205 | L-2/T-1 | 2011–12]

Magnitude Comparator

Q1. Design a magnitude comparator for comparing two 3-bit binary numbers X and Y. **(14 marks)** [CSE 205 | L-2/T-1 | 2024–25]

Q2. Design a logic circuit which finds the maximum value between two 3-bit numbers. Explain the operation using the two numbers 7 and 3. **(10 marks)** [CSE 205 | L-2/T-1 | 2020–21]

Q3. Design a logic circuit which returns the smaller value between two 4-bit binary numbers A and B. Show the operation with $A = 7$, $B = 9$. **(12 marks)** [CSE 205 | L-2/T-1 | 2018–19]

Q4. Design a digital circuit with logic gates which takes two 3-bit binary numbers X and Y and returns a 3-bit binary number Z equal to the maximum between X and Y. (X can be expressed as $X_2X_1X_0$, Y can be expressed as $Y_2XY_1Y_0$; and Z can be expressed as $Z_2Z_1Z_0$). **(10 marks)** [CSE 205 | L-2/T-1 | 2017–18]

Q5. Design and explain a 2-bit magnitude comparator. **(10 marks)** [CSE 205 | L-2/T-1 | 2016–17]

Encoders

Q1. Design a light sensor based controller circuit for accessing eight servers A, B, C, D, E, F, G, H. The light sensors are numbered as $S_1, S_2, \ldots$ and so on. The sensors of the controller will be ON to show which server is active. The servers have some priority order: C, A, E, H, D, G, F, B (C has the highest priority and B has the lowest priority). **(15 marks)** [CSE 205 | L-2/T-1 | 2024–25]

Q2. Show the truth table of an 8-to-3 line priority encoder, where higher priority is given to lower numbered input. **(9 marks)** [CSE 205 | L-2/T-1 | 2023–24]

Q3. After a school final result, Ishrat applied for college admission and gave her choices from eight colleges indexed A to H, following the priority order B, A, D, E, F, H, G, C (B has the highest priority, C has the lowest priority). Find the Boolean function of her choices and show the design steps. **(15 marks)** [CSE 205 | L-2/T-1 | 2020–21]

Q4. Design an 8-to-3 priority encoder with the priority order: 5, 0, 7, 4, 6, 3, 2, 1 (5 has highest priority, 1 has lowest priority). **(13 marks)** [CSE 205 | L-2/T-1 | 2018–19]

Q5. Design an 8-to-3 priority encoder with the priority order: 6, 0, 7, 4, 5, 3, 1, 2 (6 has highest priority, 2 has lowest priority). **(13 marks)** [CSE 205 | L-2/T-1 | 2017–18]

Q6. "A lower-order code will get priority" — Design and explain a priority encoder with negative enable for 9 bits. **(10 marks)** [CSE 205 | L-2/T-1 | 2016–17]

Q7. An Octal-to-Binary Encoder may cause ambiguities if more than one input is simultaneously set to 1. How can this ambiguity be resolved? **(4 marks)** [CSE 205 | L-2/T-1 | 2014–15]

Q8. What is a priority encoder? Design a four-input priority encoder with inputs I_0, I_1, I_2, I_3, where the priority order is $I_2 > I_3 > I_1 > I_0$. **(10 marks)** [CSE 205 | L-2/T-1 | 2013–14]

Q9. Design a 4-input priority encoder and show the gate-level circuit diagram. **(12 marks)** [CSE 205 | L-2/T-1 | 2011–12, 2013–14]

Decoders

Q1. Design a 1-bit full subtractor using a 4 to 16 line decoder that has active low outputs and two active low enables $\overline{EN_1}$ and $\overline{EN_2}$. **(12 marks)** [CSE 205 | L-2/T-1 | 2024–25]

Q2. Design a 4-to-16-line decoder by using only the minimum number of 2-to-4-line decoders. Assume that the 2-to-4-line decoders have an enable input ('1' = enabled) but the designed 4-to-16-line decoder does not have an enable input. **(12 marks)** [CSE 205 | L-2/T-1 | 2023–24]

Q3. Design a full adder circuit using a 3×8 decoder and two OR gates. **(10 marks)** [CSE 205 | L-2/T-1 | 2022–23]

Q4. Implement $F(x, y, z) = \Sigma(3, 5, 6, 7)$ using an active low 3×8 decoder and AND gates. **(10 marks)** [CSE 205 | L-2/T-1 | 2021–22]

Q5. Design a logic circuit which takes two 2-bit numbers X_1X_0 and Y_1Y_0 and finds their sum using a 2×4 line decoder with inputs I_0, I_1, active-low outputs O_3, O_2, O_1, O_0, and active-low enables $\overline{EN_1}$ and $\overline{EN_2}$. (Use the minimum number of extra logic gates.) **(15 marks)** [CSE 205 | L-2/T-1 | 2020–21]

Q6. Design a logic circuit for $f(w, x, y, z) = \Sigma(0, 2, 6, 7, 10, 11, 12, 14)$ with a 2×4 line decoder that has inputs A, B, active-low enables $\overline{EN_1}$ and $\overline{EN_2}$. (Use the minimum number of extra logic gates.) **(14 marks)** [CSE 205 | L-2/T-1 | 2018–19]

Q7. Design a logic circuit for $f(w, x, y, z) = \mathrm{SOP}(0, 2, 6, 7, 8, 10, 11, 13, 14)$ with a 2×4 line decoder that has inputs A, B, active-low enable $\overline{EN}$ and active-low outputs D'_0, D'_1, D'_2, D'_3. (Use the minimum number of extra logic gates.) **(12 marks)** [CSE 205 | L-2/T-1 | 2017–18]

Q8. Using a 2×4 line decoder, design a half subtractor. (Decoder outputs D_0, D_1, D_2, D_3 are active high; enable $\overline{EN}$ is active low.) **(10 marks)** [CSE 205 | L-2/T-1 | 2017–18]

Q9. Using only a 2×4 line decoder, solve the function $f(A, B, C, D) = \pi(3, 5, 9, 15, 12, 4, 1)$. The 2×4 line decoder has an active-low enable and active-low outputs. (Use the minimum number of gates if required.) **(10 marks)** [CSE 205 | L-2/T-1 | 2016–17]

Q10. Consider a 3×8 decoder whose outputs are active low and which has one active high enable signal. Implement the following functions using the above-mentioned decoder:

(a) $f(a,b,c,d) = \Pi\,(3,5,14)$

(b) $f(a,b,c,d) = \Sigma\,(1,6,8,13)$

Note that you will need to implement both functions in a single circuit. You may use as many decoders as required, keeping the implementation as efficient as possible. Basic gates are also allowed. **(13 marks)** [CSE 205 | L-2/T-1 | 2015–16]

Q11. Design a full adder with a decoder and basic logic gates. **(12 marks)** [CSE 205 | L-2/T-1 | 2014–15]

Q12. You need a full adder for an experiment but in the lab you could only find plenty of 3 × 8 decoders. Can any such decoder be used as a full adder? Explain your answer with a truth table. **(15 marks)** [CSE 205 | L-2/T-1 | 2012–13]

Q13. Design a 3 × 8 decoder using two 2 × 4 decoder circuits. **(8 marks)** [CSE 205 | L-2/T-1 | 2011–12]

Decoder as Demultiplexer

Q1. What is a demultiplexer? Give an example. **(8 marks)** [CSE 205 | L-2/T-1 | 2021–22]

Q2. Can a decoder act like a demultiplexer? If yes, explain how. **(5 marks)** [CSE 205 | L-2/T-1 | 2020–21]

Q3. Show that a decoder can be used as a demultiplexer. **(8 marks)** [CSE 205 | L-2/T-1 | 2018–19]

Q4. How can a decoder be used as a demultiplexer? Explain with an example. **(5 marks)** [CSE 205 | L-2/T-1 | 2016–17]

BCD to 7-Segment Decoder & BCD Error Detector

Q1. Explain how the Ripple-blanking input (RBI) and Ripple-blanking output (RBO) pins of the BCD to 7-segment decoder (IC 7447) should be connected such that all but the LSB show blank for leading zeros. **(6 marks)** [CSE 205 | L-2/T-1 | 2012–13]

Q2. You are receiving BCD codes sent by a remote transmitter. The bits are $A_3A_2A_1A_0$ (A_0 being the LSB). Design a BCD-error-detector (BED) for your receiver that will produce a HIGH for any received code that is not a valid BCD code. (For example, if you receive 1010, the output of BED will be HIGH.) **(15 marks)** [CSE 205 | L-2/T-1 | 2012–13]

Multiplexers & Demultiplexers

Q1. There is a 4×4 smart grid to serve as the electricity transmission facility. The grids will be activated based on the logic function $g(a,b,c,d) = \Sigma\,(1,2,5,7,12,14)$. Design the logic circuit using a switching circuit that has 3-bit selector bits S_2, S_1, S_0 to select any of the eight input lines. The switching circuit is activated by two Enables — one active low $\overline{EN_0}$ and another active high EN_1. (Design with the minimum number of extra logic gates if required.) **(13 marks)** [CSE 205 | L-2/T-1 | 2024–25]

Q2. Using the switching circuit defined above (5-bit selector lines A, B, C, D, E where A is MSB and E is LSB), design a 32×1 multiplexer. (Design with the minimum number of logic gates if required.) **(10 marks)** [CSE 205 | L-2/T-1 | 2024–25]

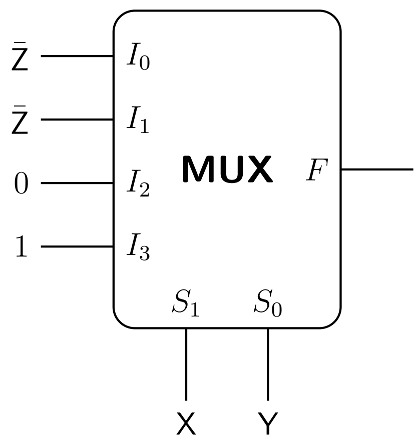
Q3. Design a full adder using NOT gates and two 4 to 1 multiplexers. **(12 marks)** [CSE 205 | L-2/T-1 | 2023–24]

Q4. Implement the following Boolean function with a 4 × 1 multiplexer and external gates:

$$F(A,B,C,D) = \Sigma\,(1,2,4,7,8,9,10,11,13,15)$$

(13 marks) [CSE 205 | L-2/T-1 | 2022–23]

Q5. What will be the output function $F(X,Y,Z)$ in the product of maxterms form in the given 4×1 MUX circuit? Assume X as the most significant bit and Z as the least significant bit. (The MUX has inputs $Z', Z, 0, 1$ and selection lines S_1S_0 connected to X, Y.)



(12 marks) [CSE 205 | L-2/T-1 | 2021–22]

Q6. Math teacher Mina Rahman sent sixteen question choices $Q_0, \ldots, Q_{15}$ to her students with a 2-bit binary code C_1C_0. If $C_1C_0 = 11$, Q_0 is selected; $C_1C_0 = 10$, Q_1 is selected; ...and so on. The code is activated by active-low $\overline{EN_1}$ and active-high EN_2. Design the multiplexing circuit. **(15 marks)** [CSE 205 | L-2/T-1 | 2020–21]

Q7. There are 16 electronic turnstile gates in a cricket stadium. The gates are numbered from 0 to 15 denoted by four bits $T_3T_2T_1T_0$. You have to design a digital ticketing system which uses a switch, and the switch has four input lines I_0, I_1, I_2, I_3. These lines can be selected by S_0, S_1. The switch is activated by two active low enables EN_1' and EN_2'. (The selection policy of the switch is as follows. If $S_1S_0 = 00$, I_0 is selected; if $S_1S_0 = 01$, I_1 is selected; if $S_1S_0 = 10$, I_2 is selected; ... so on.). You can have entry to the stadium if you get the ticket for any of the following gate numbers $(1, 3, 4, 6, 7, 9, 12, 15)$. Design the ticketing circuit using the switch. **(15 marks)** [CSE 205 | L-2/T-1 | 2020–21]

Q8. Implement $F(A, B, C, D) = \Sigma\,(1, 2, 4, 10, 12, 13, 14, 15)$ with a 4×1 multiplexer and external gates. Connect inputs A and B to the selection lines. Express F as a function of C and D for each of the four cases $AB = 00, 01, 10, 11$ to obtain the data-line requirements. **(17 marks)** [CSE 205 | L-2/T-1 | 2019–20]

Q9. Using a 4×1 line MUX, design a logic circuit for $f(w, x, y, z) = \Pi\,(0, 2, 5, 6, 7, 8, 9, 10)$, where the MUX has selector bits S_1S_0, input lines $I_0 \ldots I_3$, active-low enable $\overline{EN_1}$ and active-high enable EN_2. **(15 marks)** [CSE 205 | L-2/T-1 | 2018–19]

Q10. Design a 1-bit Full Subtractor using a 4×1 multiplexer. The MUX has selector bits S_1S_0, input lines I_0, I_1, I_2, I_3, and active-low enable $\overline{EN}$. **(12 marks)** [CSE 205 | L-2/T-1 | 2018–19]

Q11. Using an 8×1 line MUX, design a logic circuit for $f(w, x, y, z) = \mathrm{POS}(0, 2, 5, 6, 7, 8, 9, 10)$, where the MUX has selector bits $S_2S_1S_0$, input lines $I_0 \ldots I_7$, and active-low enable $\overline{EN}$. (If $S_2S_1S_0 = 000$, I_0 is selected; ...and so on.) **(15 marks)** [CSE 205 | L-2/T-1 | 2017–18]

Q12. Design a 1-bit Full Adder using two 4×1 multiplexers. The MUX has selector bits S_1S_0, input lines I_0, I_1, I_2, I_3, and active-low enable $\overline{EN}$. (If $S_1S_0 = 00$, I_0 is selected; $01 \to I_1$; $10 \to I_2$; $11 \to I_3$.) **(12 marks)** [CSE 205 | L-2/T-1 | 2017–18]

Q13. Using only 4×1 line, 2-bit multiplexers, design $f(w, x, y, z) = \pi\,(2, 4, 5, 7, 11, 12, 13)$. (Use minimum gates if required.) **(12 marks)** [CSE 205 | L-2/T-1 | 2016–17]

Q14. Design and implement the circuit diagram of a 1-bit full subtractor circuit using 2×1 MUXs only. You cannot use any other gates. **(12 marks)** [CSE 205 | L-2/T-1 | 2015–16]

Q15. Consider the Boolean function $f(w, x, y, z) = \Pi\,(2, 4, 5, 7, 11, 12, 13)$. Implement the function using a 4×1 MUX and other basic gates if necessary. **(10 marks)** [CSE 205 | L-2/T-1 | 2015–16]

Q16. Show the truth table for a three-state buffer. Construct a 2-to-1 line MUX with three-state buffers. **(3+5 marks)** [CSE 205 | L-2/T-1 | 2014–15]

Q17. An 8×1 multiplexer has inputs A, B, C, D where inputs A, B, and C are connected to selection inputs S_2, S_1, and S_0 respectively. The data inputs I_0 through I_7 are: $I_1 = I_2 = 0$; $I_3 = I_7 = 1$; $I_4 = I_5 = D$; $I_0 = I_6 = D'$. Determine the Boolean function that the multiplexer implements. **(15 marks)** [CSE 205 | L-2/T-1 | 2013–14]

Q18. Construct a 16×1 multiplexer with two 8×1 and one 2×1 multiplexers. **(5 marks)** [CSE 205 | L-2/T-1 | 2013–14]

Q19. Implement $F = \Sigma\,(3, 5, 7, 10, 11, 13, 14, 15)$ using a 16-to-1 MUX. **(10 marks)** [CSE 205 | L-2/T-1 | 2012–13]

Q20. Can the function $F = \Sigma\,(3, 5, 7, 10, 11, 13, 14, 15)$ be implemented using only one 4×1 MUX? If yes, show how; if not, explain why. **(10 marks)** [CSE 205 | L-2/T-1 | 2012–13]

Q21. Implement the following function F using one 4×1 multiplexer and basic gates:

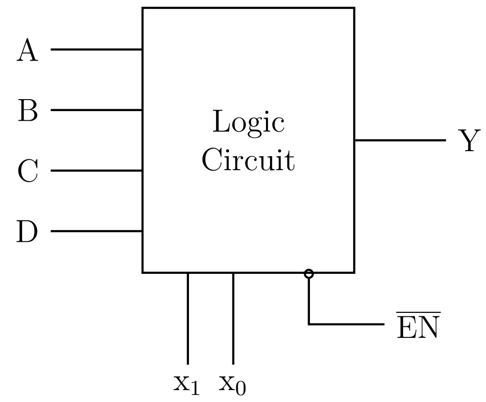
$$F(A, B, C, D) = \Sigma\,(1, 3, 4, 11, 12, 13, 14, 15)$$

(15 marks) [CSE 205 | L-2/T-1 | 2011–12]

Custom Combinational Circuit Design

Q1. Using a 4-bit binary adder and XOR gates, design a circuit which takes two 4-bit BCD inputs A and B and returns: $A + 1$ if input is 00; A' if input is 01; $A' + 1$ if input is 10; A if input is 11. **(13 marks)** [CSE 205 | L-2/T-1 | 2018–19]

Q2. Design a Binary-to-Excess-3 converter using a logic circuit block with inputs A, B, C, D, control lines x_1, x_0, and active-low $\overline{EN}$. Output Y: if $x_1x_0 = 00$, $A \to Y$; if 01, $B \to Y$; if 10, $C \to Y$; if 11, $D \to Y$.



(8 marks) **[CSE 205 | L-2/T-1 | 2016–17]**

Q3. Design a combinational circuit with 3 inputs x, y, z and 3 outputs A, B, C. When the binary input is 0, 1, 2, or 3, the binary output is two greater than the input. When the binary input is 4, 5, 6, or 7, the binary output is three less than the input. **(18 marks)** **[CSE 205 | L-2/T-1 | 2014–15]**

BUET · CSE 205 · Digital Logic Design

Digital Logic Design

Flip-Flops, Latches & Race-Around Problems

SR, JK, D, T Flip-Flops, Excitation Tables, Race Conditions

S5 Flip-Flops, Latches & Race-Around Problems

SR Latch & SR/JK Undefined Condition

Q1. Discuss why the condition $S = R = 1$ leads to an unstable condition for an SR latch. **(7 marks)** [CSE 205 | L-2/T-1 | 2018–19]

Q2. Consider the following transition table:

$y \setminus x_1x_2$	00	01	11	10
0	0	0	1	0
1	0	1	1	1

Design the circuit with NAND SR latches and gates. **(10 marks)** [CSE 205 | L-2/T-1 | 2016–17]

Q3. Write the excitation table for a SR latch constructed with NOR gates. **(4 marks)** [CSE 205 | L-2/T-1 | 2015–16]

Q4. What do you understand by mechanical switch contact bounce? Describe how the $\overline{S}\,\overline{R}$ latch can eliminate mechanical switch contact bounce. **(10 marks)** [CSE 205 | L-2/T-1 | 2014–15]

Q5. Explain the undefined condition in SR flip-flops. How is this problem solved in JK flip-flops? **(4+4 marks)** [CSE 205 | L-2/T-1 | 2012–13]

Q6. Show the logic diagram and function table of an SR latch with NAND gates. **(4 marks)** [CSE 205 | L-2/T-1 | 2011–12]

JK Flip-Flop Characteristics

Q1. What is the problem in the construction of a JK flip-flop? How can we overcome this problem? **(6 marks)** [CSE 205 | L-2/T-1 | 2023–24]

Q2. Derive the characteristic table and the characteristic equation for a JK flip-flop. **(4 marks)** [CSE 205 | L-2/T-1 | 2011–12]

Master-Slave D Flip-Flop

Q1. A clocked sequential circuit uses a master-slave configuration.

(a) Explain how signal propagation through the master and slave stages affects the timing behavior of the circuit.

(b) Analyze and justify the possibility of race-around under non-ideal clock conditions (finite rise and fall time).

(12 marks) [CSE 205 | L-2/T-1 | 2024–25]

Q2. Draw the block diagram of a negative edge triggered Master-Slave D flip-flop using D latches. **(10 marks)** [CSE 205 | L-2/T-1 | 2022–23]

Q3. Draw the circuit diagram of a master-slave JK flip-flop with asynchronous preset and clear using only NOR gates. **(5 marks)** [CSE 205 | L-2/T-1 | 2021–22]

Q4. Draw the circuit diagram of a master-slave JK Flip-Flop using only NAND gates. **(8 marks)** [CSE 205 | L-2/T-1 | 2018–19]

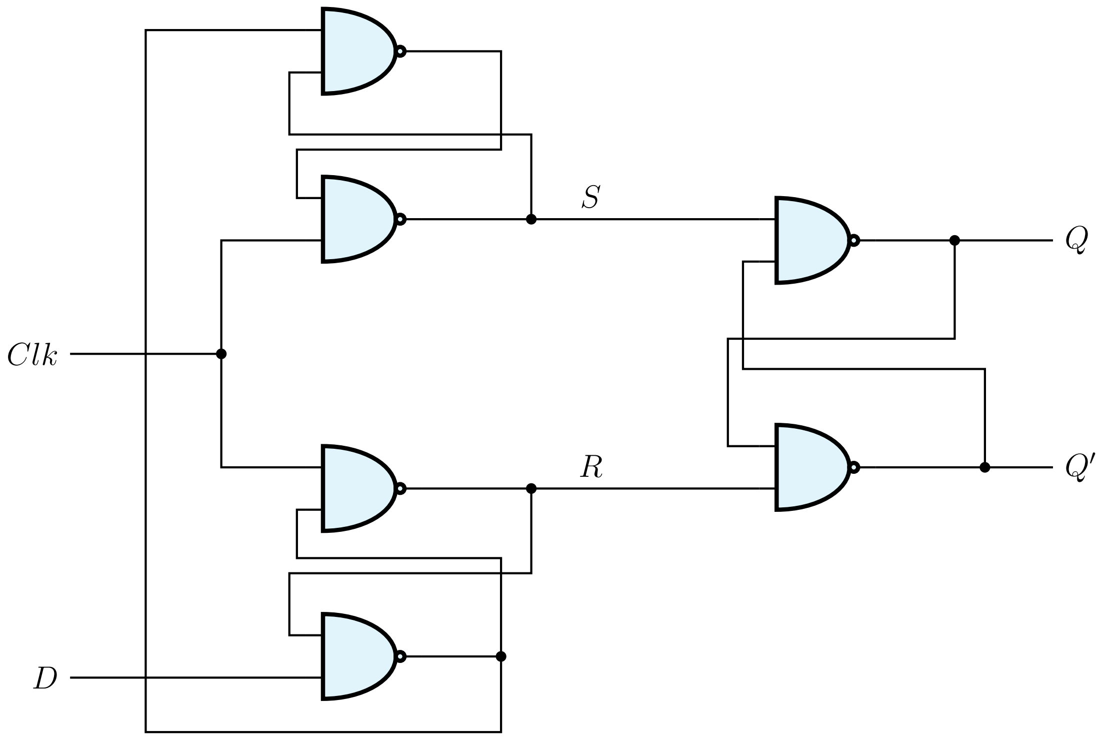
Q5. Why do we use master-slave flip-flops? **(5 marks)** [CSE 205 | L-2/T-1 | 2017–18]

Q6. Draw the block diagram and explain the operation of a Master-Slave negative edge triggered D flip-flop. **(12 marks)** [CSE 205 | L-2/T-1 | 2015–16]

Q7. Draw and explain the operation of a Master-Slave D flip-flop. **(12 marks)** [CSE 205 | L-2/T-1 | 2011–12]

Sequential Circuit Timing Analysis

Q1. Consider a sequential circuit. Let the current values be: CLK = 1, $D = 1$, $Q = 1$, $Q' = 0$. If the value of D changes from 1 to 0 after the "Hold Time" ends and CLK remains unchanged at 1, what will be the values of Q and Q'?



(6 marks) [CSE 205 | L-2/T-1 | 2015–16]

Latch vs Flip-Flop

Q1. Write the differences between a latch and a flip-flop. **(5 marks)** [CSE 205 | L-2/T-1 | 2013–14]

Flip-Flop Conversion

Q1. Design a T flip-flop using a JK flip-flop and verify its operation using state transitions. **(10 marks)** [CSE 205 | L-2/T-1 | 2024–25]

Q2. Construct a JK flip-flop from a T flip-flop and necessary gates. Show all the steps of this construction process. **(11 marks)** [CSE 205 | L-2/T-1 | 2023–24, 2022–23, 2016–17]

Q3. Design a D flip-flop using a T flip-flop and gates. **(5 marks)** [CSE 205 | L-2/T-1 | 2020–21]
↪ *Similar: 2019–20*

Q4. Design a SR Flip-Flop using a D Flip-Flop and a 4×1 MUX. **(10 marks)** [CSE 205 | L-2/T-1 | 2014–15]

Q5. Convert: (i) a JK flip-flop into a T flip-flop, and (ii) a D flip-flop into a JK flip-flop. **(8 marks)** [CSE 205 | L-2/T-1 | 2013–14]

Q6. Convert a T flip-flop into a JK flip-flop and vice versa. **(4+4 marks)** [CSE 205 | L-2/T-1 | 2012–13]

Q7. Convert a D flip-flop to a T flip-flop. Use necessary gates. **(4 marks)** [CSE 205 | L-2/T-1 | 2011–12]

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Digital Logic Design

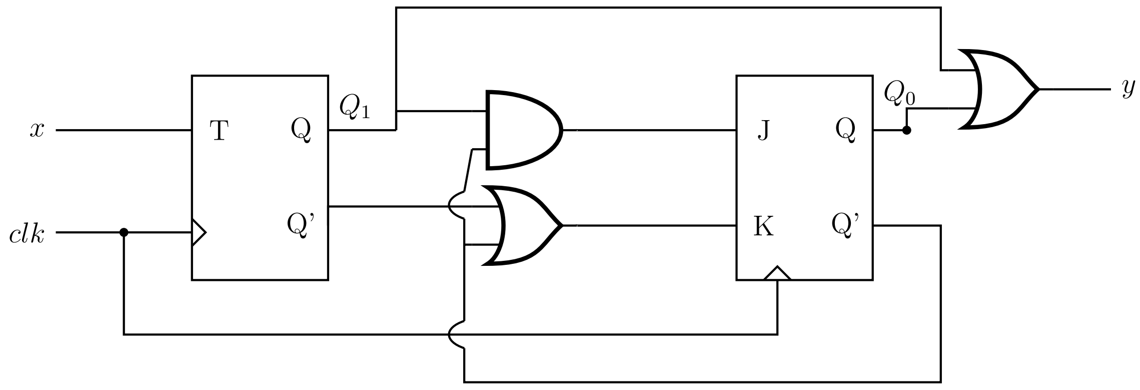
Sequential Circuits, State Diagrams & FSM

Mealy & Moore Machines, State Tables, State Minimization, Pulse/Fundamental Mode

S6 Sequential Circuits, State Diagrams & FSM

State Diagram Derivation from Circuit

Q1. Analyze the sequential circuit in the figure below. Derive the transition table, excitation table and state table.



(12 marks) [CSE 205 | L-2/T-1 | 2023–24]

Q2. A sequential circuit has two flip-flops A and B, two inputs x_1 and x_2, and one output z. The flip-flop input equations and circuit output equations are:

$$J_A = y_A x_1 + y_B x_2, \qquad K_A = y_A y_B + x_1 x_2$$
$$J_B = y_A x_1, \qquad K_B = y_A y_B x_1 + y_B x_2$$
$$z = y_A x_1 x_2 + y_B x_1 x_2$$

(a) Draw the logic diagram of the circuit.

(b) Derive the transition table, excitation table and state table.

(c) Draw the state diagram.

(20 marks) [CSE 205 | L-2/T-1 | 2021–22, 2019–20]

Q3. A sequential circuit has two JK Flip-Flops A and B, two inputs x and y, and one output z. The flip-flop input equations and circuit output equation are:

$$J_A = Ax + By, \qquad K_A = AB + xy$$
$$J_B = Ax, \qquad K_B = ABx + By, \qquad z = Axy + Bxy$$

(a) Draw the logic diagram of the circuit.

(b) Derive the state table.

(c) Draw the state diagram.

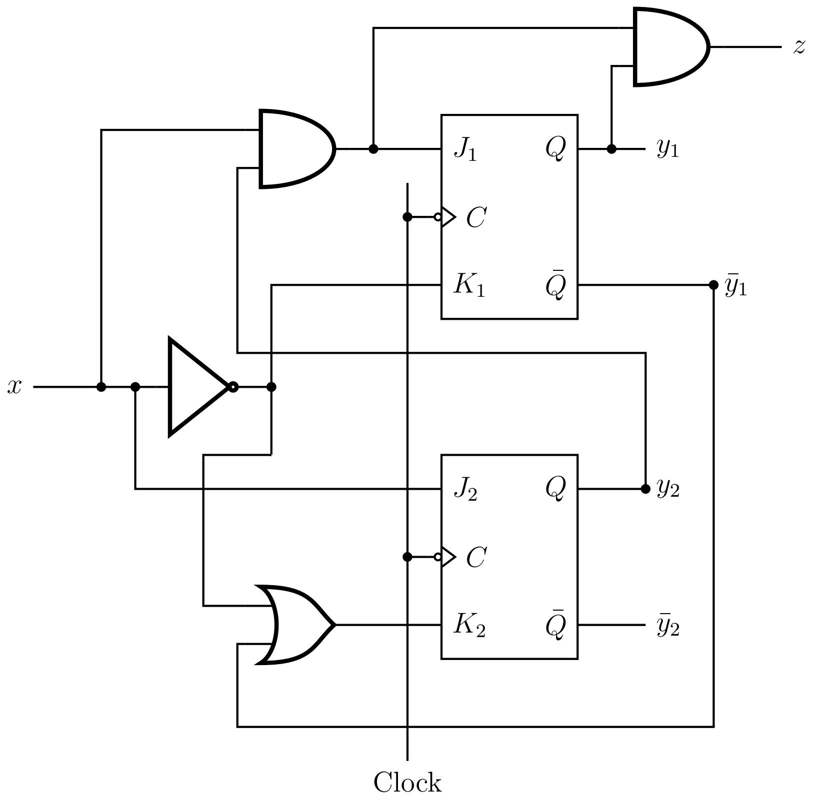
(4+7+4 marks) [CSE 205 | L-2/T-1 | 2016–17]

Q4. A sequential circuit has two JK Flip-Flops A and B, two inputs x and y, and one output z. The Flip-Flop input equations and circuit output equation are:

$$J_A = Bx + B'y', \quad K_A = B'xy', \quad J_B = A'x, \quad K_B = A + xy', \quad z = Ax'y' + Bx'y'$$

Draw the logic diagram of the circuit, tabulate the state table and derive the state equations for A and B. **(15 marks)** [CSE 205 | L-2/T-1 | 2014–15]

Q5. Derive the state diagram from the following sequential circuit.



(15 marks) [CSE 205 | L-2/T-1 | 2012–13]

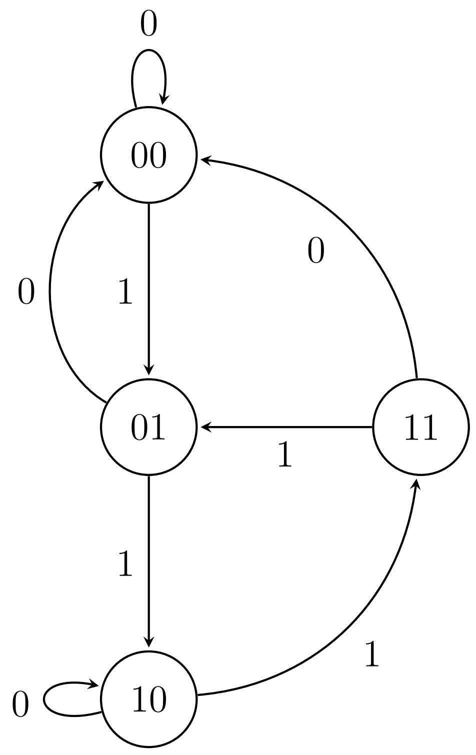
Circuit Design from State Diagram

Q1. Design the circuit corresponding to the given state diagram using T flip-flops:

(a) Write the state table.

(b) Find the input equations for the T flip-flops.

(c) Draw the circuit diagram.



(21 marks) [CSE 205 | L-2/T-1 | 2013–14]

Q2. Design a one-input, one-output serial 2's complementer using D flip-flops. The circuit accepts a string of bits from the input and generates the 2's complement at the output. The circuit can be reset asynchronously to start and end the operation. Include: (i) State diagram, (ii) State table, (iii) Simplified logic equations, (iv) Logic diagram. **(15 marks)** [CSE 205 | L-2/T-1 | 2015–16]

Q3. Design a control unit with a synchronous sequential circuit for a vending machine that accepts 5 Tk, 10 Tk and 20 Tk notes and releases a soft drink bottle worth 20 Tk. The machine releases a drink whenever the deposit amount equals or exceeds 20 Tk and keeps the change. The maximum deposit at any time is 15 Tk. The machine releases the deposit whenever the 'CHANGE' input is given. Outputs: Release Drink Z_1 and Release Change Z_2. Inputs X_1X_2: Tk5=00, Tk10=01, Tk20=10, CHANGE=11. Use JK Flip-Flops.

Input	X_1X_2
Tk 5	00
Tk 10	01
Tk 20	10
CHANGE	11

(20 marks) [CSE 205 | L-2/T-1 | 2020–21]

Sequence Detector

Q1. Develop a sequential circuit that detects a sequence of three or more consecutive 1's in a string of bits coming through an input line. Use D flip-flops. Show the state diagram, the state table and simplified logic equations. You do not need to show the logic diagram. **(12 marks)** [CSE 205 | L-2/T-1 | 2022–23]

Q2. Design a sequence recognizer using T flip-flops that recognizes a 4-bit **non-overlapping** binary sequence X from an input stream of bits, where:

$$\text{Sequence } X = \text{4-bit binary representation of your roll number} \pmod{16}$$

(25 marks) [CSE 205 | L-2/T-1 | 2019–20]

Q3. Design a sequence recognizer that recognizes the sequence **1101** from an input sequence (overlapping allowed). Design the circuit using SR Flip-Flops and draw the circuit diagram. **(20 marks)** [CSE 205 | L-2/T-1 | 2018–19]

Q4. Design a clocked sequential circuit that recognizes the input sequence **1010** (including overlap) such that for input $x = 01010100011010010100$ the output is $z = 00001010000001000010$. Design the circuit using T Flip-Flops and basic gates. **(20 marks)** [CSE 205 | L-2/T-1 | 2016–17]

Q5. Draw the state diagram in Moore model of a synchronous sequential circuit that results in an output of 1 whenever the sequence 10011 occurs. The circuit is required to recognize overlapping sequences. **(8 marks)** [CSE 205 | L-2/T-1 | 2012–13]

Q6. Design a sequential circuit that detects a sequence of three or more consecutive 1s in a string of bits coming through an input line. Use D flip-flops. The design must include the following:

(a) State diagram
(b) State table
(c) Simplified logic equations
(d) Logic

(15 marks) [CSE 205 | L-2/T-1 | 2011–12]

Mealy & Moore Model Comparison & Block Diagrams

Q1. Draw the block diagram of Mealy and Moore state machines. State the difference between two models with a simple example. **(10 marks)** [CSE 205 | L-2/T-1 | 2024–25, 2022–23, 2015–16, 2013–14, 2011–12]

Q2. Transform the Mealy machine shown below into a corresponding Moore machine.

PS	NS, Z	
	$x = 0$	$x = 1$
A	B, 1	C, 0
B	C, 1	D, 0
C	D, 0	E, 1
D	E, 0	A, 1
E	A, 1	B, 1

(6 marks) [CSE 205 | L-2/T-1 | 2023–24]

Q3. Transform the following Mealy machine into a corresponding Moore machine.

PS	NS, Z	
	x = 0	x = 1
A	C, 0	B, 0
B	A, 1	D, 0
C	B, 1	A, 1
D	D, 1	C, 0

(5 marks) [CSE 205 | L-2/T-1 | 2021–22]

Q4. Obtain the Mealy equivalent state table for the following Moore machine.

PS	NS		Z
	x = 0	x = 1	
A	A	B	0
B	C	B	0
C	C	D	1
D	A	D	0

(7 marks) [CSE 205 | L-2/T-1 | 2018–19]

Q5. Define (i) a Mealy machine and (ii) a Moore machine. **(5 marks)** [CSE 205 | L-2/T-1 | 2017–18]

Q6. What are the advantages and disadvantages of using the Mealy model and the Moore model for sequential circuit design? **(6 marks)** [CSE 205 | L-2/T-1 | 2012–13]

State Minimization (Implication Table)

Q1. For the state-table of a completely specified circuit (PS: A–H), find the equivalence partitions and write the state-table of the minimal machine. Name the state that are equivalent .

PS	NS, Z $(x = 0)$	NS, Z $(x = 1)$
A	B,1	H,1
B	F,1	D,1
C	D,0	E,1
D	C,0	F,1
E	D,1	C,1
F	C,1	C,1
G	C,1	D,1
H	C,0	A,1

(10 marks) [CSE 205 | L-2/T-1 | 2020–21]

Q2. For the state-table of a completely specified circuit, find the equivalent partitions and write the state-table of the minimal machine. Name the equivalent states.

PS	NS, Z		
	I	J	K
A	A,0	B,1	E,1
B	B,0	A,1	F,1
C	A,1	D,0	E,0
D	F,0	C,1	A,0
E	A,0	D,1	E,1
F	B,0	D,1	F,1

(8 marks) [CSE 205 | L-2/T-1 | 2018–19]

Q3. For the state-table of a completely specified circuit, find the equivalence partitions and write the state-table of the minimal machine. Name the equivalent states.

x_1x_2	NS,Z			
PS	00	01	11	10
A	D, 0	D, 0	F, 0	A, 0
B	C, 1	D, 0	E, 1	F, 0
C	C, 1	D, 0	E, 1	A, 0
D	D, 0	B, 0	A, 0	F, 0
E	C, 1	F, 0	E, 1	A, 0
F	D, 0	D, 0	A, 0	F, 0
G	G, 0	G, 0	A, 0	A, 0
H	B, 1	D, 0	E, 1	A, 0

(10 marks) [CSE 205 | L-2/T-1 | 2017–18]

Q4. You are given the following state table of a synchronous sequential circuit.

Present State	Next State		Output	
	$x = 0$	$x = 1$	$x = 0$	$x = 1$
a	a	c	0	0
b	d	a	1	0
c	f	f	0	0
d	e	b	1	0
e	g	g	1	0
f	c	c	0	0
g	b	h	1	0
h	h	c	0	0

(i) Find the reduced state table using an implication table.

(ii) Draw the equivalent minimized state diagram.

(10 marks) [CSE 205 | L-2/T-1 | 2016–17]

Q5. You are given a state table of a synchronous sequential circuit. Find the reduced state table using the implication table.

Present State	Next State		Output	
	x=0	x=1	x=0	x=1
a	d	b	0	0
b	e	a	0	0
c	g	f	0	1
d	a	d	1	0
e	a	d	1	0
f	c	b	0	0
g	a	e	1	0

(12 marks) [CSE 205 | L-2/T-1 | 2015–16]

Q6. Find a critical race-free state assignment for a reduced flow table (states *a*–*d*, inputs 00/01/11/10) using the Multiple-row method.

	00	01	11	10
a	a	c	a	d
b	a	b	c	b
c	c	c	c	d
d	d	b	a	d

(10 marks) [CSE 205 | L-2/T-1 | 2014–15]

Q7. Using an implication table, reduce the following state table to a minimum number of states and write the reduced state table.

Present State	Next State		Output	
	$x = 0$	$x = 1$	$x = 0$	$x = 1$
A	F	B	0	0
B	D	C	0	0
C	F	E	0	0
D	G	A	1	0
E	D	C	0	0
F	F	B	1	1
G	G	H	0	1
H	G	A	1	0

(15 marks) [CSE 205 | L-2/T-1 | 2013–14]

Q8. Using an implication table, reduce the following state table to a minimum number of states and write the reduced state table.

	X	
	0	**1**
A	E/0	D/0
B	A/1	F/0
C	C/0	A/1
D	B/0	A/0
E	D/1	C/0
F	C/0	D/1
G	H/1	G/1
H	C/1	B/1

(17 marks) [CSE 205 | L-2/T-1 | 2012–13]

Incompletely Specified Machine Minimization

Q1. For the state table of a completely specified asynchronous sequential circuit, find out the equivalent partitions, implication table. Formulate the state table of the minimal machine. Find out the states those are equivalent.

x1x2	NS, z			
PS	x1x2=00	x1x2=01	x1x2=11	x1x2=10
A	D/0	D/0	F/0	A/0
B	C/1	D/0	E/1	F/0
C	C/1	D/0	E/1	A/0
D	D/0	B/0	A/0	F/0
E	C/1	F/0	E/1	A/0
F	D/0	D/0	A/0	F/0
G	G/0	G/0	A/0	A/0
H	B/1	D/0	E/1	A/0

(PS = Present state, NS = next state, x1, x2 are input, z is output)

(12 marks) [CSE 205 | L-2/T-1 | 2024–25]

Q2. For the incompletely specified machine shown below, derive the implication table and determine the maximal compatibles and the maximal incompatibles. Find the upper bound and the lower bound of the number of states of the minimal machine. Give the state table of the minimal machine.

PS	NS, Z	
	$x = 0$	$x = 1$
A	B, 0	C, 0
B	D, 0	–, –
C	F, 0	E, 1
D	B, 0	G, 0
E	F, 0	C, 1
F	E, 0	D, 1
G	F, 0	–, –

(23 marks) [CSE 205 | L-2/T-1 | 2023–24]

Q3. Reduce the number of states of the following state table using implication table. Show the reduced state table.

State	Next State		Output	
	Input X = 0	Input X = 1	Input X = 0	Input X = 1
A	A	B	0	0
B	C	D	0	1
C	A	B	0	0
D	E	D	0	1
E	A	D	0	1

(11 marks) [CSE 205 | L-2/T-1 | 2022–23]

Q4. For the incompletely specified machine shown in the figure, derive the implication table and determine the maximal compatibles and the maximal incompatibles. Find the upper bound and the lower bound of the number of states of the minimal machine. Give the state table of the minimal machine.

PS	NS, Z		
	I_1	I_2	I_3
A	C, 0	E, 1	-
B	C, 0	E, -	-
C	B, -	C, 0	A, -
D	B, 0	C, -	E, -
E	-	E, 0	A, -

(15 marks) [CSE 205 | L-2/T-1 | 2021–22]

Q5. For the incompletely specified machine below, complete the implication table and determine the maximal compatibles and maximal incompatibles. Find the upper and lower bounds on the number of states of the minimal machine. Give the state table of the minimal machine.

PS	NS, Z_1Z_2			
	00	01	11	10
A	A, 00	E, 01	-	A, 01
B	-	C, 10	B, 00	D, 11
C	A, 00	C, 10	-	-
D	A, 00	-	-	D, 11
E	-	E, 01	F, 00	-
F	-	G, 10	F, 00	G, 11
G	A, 00	-	-	G, 11

(25 marks) [CSE 205 | L-2/T-1 | 2020–21]

Q6. For the incompletely specified machine shown below, derive the implication table and determine the maximal compatibles and the maximal incompatibles. Find the upper bound and the lower bound of the minimal machine and determine the minimal machine.

PS	NS, Z	
	x = 0	x = 1
A	A, -	B, 1
B	G, -	D, 0
C	B, 1	B, -
D	A, 1	B, -
E	C, -	A, -
F	F, -	C, -
G	G, -	G, -

(25 marks) [CSE 205 | L-2/T-1 | 2019–20]

Q7. For the incompletely specified machine , complete the implication table and determine the maximal compatibles and maximal incompatibles. Find the upper and lower bounds. Give the state table of the minimal machine.

PS	NS, Z	
	x = 0	x = 1
A	B, 0	C, 1
B	D, 0	C, 1
C	A, 0	E, 0
D	-	F, 1
E	G, 1	F, 0
F	B, 0	-
G	D, 0	E, 0

(20 marks) [CSE 205 | L-2/T-1 | 2018–19]

Q8. For the incompletely specified machine , complete the implication table and determine the maximal compatibles and maximal incompatibles. Find the upper and lower bounds on the number of states of the minimal machine. Give the state table of the minimal machine.

PS	NS,Z	
	x = 0	x = 1
A	A, -	B, 1
B	G, -	D, 0
C	B, 1	B, -
D	A, 1	B, -
E	C, -	A, -
F	F, -	C, -
G	G, -	G, -

(15 marks) [CSE 205 | L-2/T-1 | 2017–18]

Redundant States

Q1. What are the problems that may occur when redundant states are present in the state diagram? **(6 marks)** [CSE 205 | L-2/T-1 | 2012–13, 2013–14]

Output Assignment in Flow Table

Q1. Write down the procedure to assign the output to unstable states in a flow table. **(7 marks)** [CSE 205 | L-2/T-1 | 2011–12]

Fundamental Mode Design

Q1. Design an asynchronous circuit functioning in fundamental mode. The circuit has three inputs a, b, and c, and one output U. $U = 1$ only if all inputs are equal to '1', and they went to '1' in the order a, b, then c. U returns to 0 when all inputs are returned to '0'. Derive the primitive flow table, reduced flow table and the minimum number of states required to have a race-free state assignment, and find a possible state assignment. **(23 marks)** [CSE 205 | L-2/T-1 | 2023–24]

Q2. At a junction of a single-track railroad and a road, traffic lights are to be installed. The lights are to be controlled by switches that are pressed or released by the trains. When a train approaches the junction from either direction and is within 1500 feet from it, the lights are to change from green to red and remain red until the train is 1500 feet past the junction. You may assume that the length of a train is smaller than 3000 feet. Determine a primitive flow table and reduce it. **(20 marks)** [CSE 205 | L-2/T-1 | 2021–22]

Q3. Design a two-input (x_1, x_2), one-output (z) fundamental mode circuit that operates as follows. The output changes from 0 to 1 only on the first x_1 input change that follows an x_2 input change. A $1 \to 0$ output change occurs only when x_1 changes from 1 to 0 while $x_2 = 1$. Determine the primitive flow table, reduced flow table, race-free state assignment, K-maps and circuit diagram. **(25 marks)** [CSE 205 | L-2/T-1 | 2020–21]

Q4. Design a fundamental mode circuit to function as an electrical lock. The lock has two switch inputs X_1 and X_2. Design the circuit so that the lock-open signal $Z = 1$ is produced only after the following conditions have been satisfied:

(a) Begin with $X_1 = X_2 = 0$.

(b) While $X_2 = 0$, X_1 is turned on and then turned off.

(c) Then while X_1 remains off, X_2 is turned on to open the lock ($Z = 1$).

(d) For any deviation in the sequence of the above transitions, the lock remains closed ($Z = 0$).

Determine the primitive flow table, reduced flow table, race-free state assignment, K-maps and circuit diagram. **(25 marks)** [CSE 205 | L-2/T-1 | 2019–20]

Q5. Construct a primitive flow table for a fundamental mode circuit where the output $z = 0$ when the two inputs x_1 and x_2 are equal. $z = 1$ results when $x_1 = 0$ and x_2 changes from 0 to 1, and when $x_1 = 1$ and x_2 changes from 1 to 0. No other input change causes any output change. Determine the reduced flow table containing the minimum number of rows. **(10+10 marks)** [CSE 205 | L-2/T-1 | 2018–19]

Q6. Design a circuit for a digital lock using T latches. The lock has two push-button inputs A and B and produces a single pulse Z_1 for each activation. The two switches are mechanically interlocked so that simultaneous pulses are not possible. Features: (i) opening sequence is AABABA; (ii) three B pulses give absolute reset; (iii) any incorrect use of A produces output Z_2 to ring a bell. **(20 marks)** [CSE 205 | L-2/T-1 | 2018–19]

Q7. Construct a primitive flow table for a fundamental mode circuit with two inputs (x_1, x_2) and two outputs (z_1, z_2). Specifications: when $x_1x_2 = 00$, $z_1z_2 = 00$. Output 10 is produced following input sequence $00 \to 01 \to 11$ and remains until x_1x_2 returns to 00. Output 01 is produced following sequence $00 \to 10 \to 11$ and remains until 00 input returns the output to 00. Determine the minimum row-reduced flow table. **(10+10 marks)** [CSE 205 | L-2/T-1 | 2017–18]

Q8. What does it mean when an asynchronous sequential circuit is assumed to operate in the fundamental mode? **(4 marks)** [CSE 205 | L-2/T-1 | 2015–16]

Q9. Derive the primitive flow table for a negative-edge triggered T flip-flop. **(12 marks)** [CSE 205 | L-2/T-1 | 2015–16]

Q10. A specification is given below for a gated latch circuit with two inputs G and D and an output Q. Assume fundamental mode operation.

"Binary information present at the D input is transferred to the Q output when G is equal to 1. The Q output will follow the D input as long as G = 1. When G goes to 0, the information that was present at the D input at the time the transition occurred is retained at the Q output."

Derive the primitive flow table and then obtain the reduced flow table. **(6+6 marks)** [CSE 205 | L-2/T-1 | 2011–12]

Excitation Function & Transition Table

Q1. An asynchronous sequential circuit is described by the excitation function $Y = x_1x_2' + (x_1 + x_2')y$ and the output function $z = y$.

(a) Draw the logic diagram of the circuit.

(b) Derive the transition table and output map.

(c) Obtain a two-state flow table.

(12 marks) [CSE 205 | L-2/T-1 | 2022–23]

Q2. An asynchronous sequential circuit has two internal states and one output. The excitation and output functions are:

$$\begin{aligned} Y_1 &= x_1x_2 + x_1y_2' + x_2'y_1 \\ Y_2 &= x_2 + x_1y_1'y_2 + x_1'y_1 \\ Z &= x_2 + y_1 \end{aligned}$$

Draw the logic diagram of the circuit and derive the transition table. **(15 marks)** [CSE 205 | L-2/T-1 | 2016–17]

Q3. An asynchronous sequential circuit is described by the following excitation function:

$$Y = X_1X_2' + (X_1 + X_2')\,y$$

Draw the logic diagram of the circuit and derive the transition table. **(10 marks)** [CSE 205 | L-2/T-1 | 2015–16]

Hazards in Asynchronous Circuits

Q1. What is a static-1 hazard? Give an example of a static-1 hazard in an asynchronous sequential circuit. **(5 marks)** [CSE 205 | L-2/T-1 | 2015–16]

Serial Parity Generation

Q1. Design a serial parity generation circuit. The circuit receives a sequence of bits and determines whether the sequence contains an even or odd number of ones. The circuit output p should be 0 for even parity and 1 for odd parity. **(10 marks)** [CSE 205 | L-2/T-1 | 2020–21, 2018–19]

Q2. For the following primitive flow table shown in Figure , answer the following questions:

(i) Find all compatible pairs by means of an implication table.

(ii) Find the maximal compatibles by means of a merger diagram.

(iii) Find a minimal set of compatibles that covers all the states and is closed.

	00	01	11	10
a	a, 0	b, –	–, –	e, –
b	a, –	b, 0	c, –	–, –
c	–, –	d, –	c, 0	h, –
d	a, –	d, 1	–, –	–, –
e	a, –	–, –	f, –	e, 0
f	–, –	g, –	f, 0	h, –
g	a, –	g, 0	–, –	–, –
h	a, –	–, –	–, –	h, 0

(10+5+5 marks) [CSE 205 | L-2/T-1 | 2014–15]

Race Around Condition

Q1. What is a race in fundamental mode circuits? Explain critical and non-critical races with examples. **(12 marks)** [CSE 205 | L-2/T-1 | 2022–23]

Q2. For the given reduced flow table, find a valid assignment without any critical race and complete the modified flow table.

	00	01	11	10
A	(A),0	D,0	B,-	(A),0
B	A,-	-	C,-	-
C	A,-	(C),1	(C),1	(C),1
D	A,0	(D),0	(D),0	C,-

(10 marks) [CSE 205 | L-2/T-1 | 2021–22]

Q3. For the given reduced flow table, find a valid assignment without any critical race and complete the modified flow table.

x_1x_2	00	01	11	10
a	a/0	b/-	a/1	b/-
b	a/-	b/0	c/-	b/0
c	a/-	c/1	c/0	b/-

(10 marks) [CSE 205 | L-2/T-1 | 2020–21]

Q4. For the given reduced flow table, find a valid assignment without any critical race and complete the modified flow table.

x_1x_2	00	01	11	10
a	a/0	d/0	a/1	c/0
b	b/0	c/-	b/0	d/-
c	b/0	c/1	a/1	c/0
d	a/0	d/0	b/0	d/1

(8 marks) [CSE 205 | L-2/T-1 | 2018–19]

Q5. For the given reduced flow table, find a valid assignment without any critical race and complete the modified flow table.

x_1x_2	00	01	11	10
a	a/0	d/0	a/1	c/0
b	b/0	c/-	b/0	d/-
c	b/0	c/1	a/1	c/0
d	a/0	d/0	b/0	d/1

(10 marks) [CSE 205 | L-2/T-1 | 2017–18]

Q6. What is race condition? Explain different types of races with examples. **(10 marks)** [CSE 205 | L-2/T-1 | 2016–17]

Q7. Investigate the transition table (state variables y_1y_2, inputs x_1x_2) and determine all race conditions and whether they are critical or noncritical. Determine also whether there are any cycles.

y_1y_2 \ x_1x_2	00	01	11	10
00	10	00	11	10
01	01	00	10	10
11	01	00	11	11
10	11	00	10	10

(10 marks) [CSE 205 | L-2/T-1 | 2014–15]

Q8. What is a race in fundamental mode circuits? Explain critical and non-critical races with an example. **(4+8 marks)** [CSE 205 | L-2/T-1 | 2011–12]

Pulse Mode Logic

Q1. Design an automatic vending machine, where the candy bars inside the machine cost 3 Tk and the machine accepts 1 Tk and 2 Tk only. The system accepts the bills sequentially. The circuit produces a level output that delivers the candy whenever the amount received by the machine is 3 Tk or more. The excess amount is kept deposited for the next candy. Design a pulse mode circuit for the vending machine using S-R flip-flops. Provide the circuit diagram as well. **(25 marks)** [CSE 205 | L-2/T-1 | 2024–25]

Q2. Design a pulse mode sequential circuit that satisfies the following requirements. The circuit has two input lines x_1 and x_2 along with one level output line z. An output transition from 0 to 1 is produced only after the occurrence of the last x_2 pulse in the sequence x_1-x_2-x_1-x_2. The output is reset from 1 to 0 only after the first x_1 pulse that occurs following the 0 to 1 output transition. The circuit allows overlapping sequences. Develop the pulse mode circuit using JK flip-flops. Draw the circuit diagram. **(23 marks)** [CSE 205 | L-2/T-1 | 2023–24]

Q3. Design an automatic vending machine. The candy bars inside the machine cost 15 Tk, and the machine accepts 5 Tk and 10 Tk only. An electromechanical system accepts the money sequentially. The circuit produces a pulse output that delivers the candy whenever the amount received by the machine is 15 Tk or more. The excess amount is kept deposited for the next candy. Design a pulse mode circuit for the vending machine using S-R latches. **(20 marks)** **[CSE 205 | L-2/T-1 | 2022–23]**

Q4. Design a circuit for a digit lock using T latches. The lock has two input push-button switches A, B and it outputs a single pulse (Z_1) for each activation. The two switches are interlocked mechanically so that simultaneous pulses are not possible. The lock should have the following features: (i) The opening sequence is AABABA. (ii) Three B pulses should give an absolute reset. (iii) Any incorrect use of the A switch will cause an output (Z_2) to ring a bell to warn that the lock is being tampered with. **(20 marks)** **[CSE 205 | L-2/T-1 | 2021–22]**

Q5. Design an automatic vending machine. Candy bars cost 3 Tk; the machine accepts 1 Tk and 2 Tk coins sequentially. The circuit produces a level output delivering the candy whenever the amount received equals or exceeds 3 Tk. Excess amount is kept for the next candy. Design a pulse mode circuit using SR Flip-Flops. Draw the circuit diagram. **(20 marks)** **[CSE 205 | L-2/T-1 | 2017–18]**

Q6. Write down the restrictions on the duration of input for pulse mode asynchronous circuits. **(4 marks)** **[CSE 205 | L-2/T-1 | 2011–12]**

BUET · CSE 205 · Digital Logic Design

Digital Logic Design

Registers & Counters

Shift Registers, Asynchronous & Synchronous Counters, Ring Counters

S7 Registers & Counters

Ripple (Asynchronous) Counter

Q1. Design the Modulo-100 ripple down counter with JK flip-flops and basic gates. **(6 marks)** [CSE 205 | L-2/T-1 | 2023–24]

Q2. Show the logic diagram of a four-bit binary ripple (asynchronous) countdown counter using T flip-flops that trigger on the negative edge of the clock. **(12 marks)** [CSE 205 | L-2/T-1 | 2022–23]

Q3. A MOD-13 counter counts from 0000 to 1100. Design the MOD-13 ripple up/down counter with positive edge-triggered JK flip-flops and basic gates. **(5 marks)** [CSE 205 | L-2/T-1 | 2021–22]

Q4. Design a BCD ripple down counter using D flip-flops. **(5 marks)** [CSE 205 | L-2/T-1 | 2020–21, 2018–19]

Q5. Design a 4-bit modulo-12 asynchronous up counter using JK Flip-Flops. **(5 marks)** [CSE 205 | L-2/T-1 | 2017–18]

Q6. A MOD-12 counter counts from 0000 to 1011. Design the MOD-12 ripple counter with positive edge-triggered JK Flip-Flops and basic gates. **(10 marks)** [CSE 205 | L-2/T-1 | 2016–17]

↪ *Similar: 2019–20*

Q7. Design a 4-bit Ripple counter using D Flip-Flops and briefly explain its operation. **(10 marks)** [CSE 205 | L-2/T-1 | 2014–15]

Q8. (a) Design a modulo-6 3-bit asynchronous (ripple) up counter with positive edge-triggered JK flip-flops.

(b) Show the timing diagram for the above designed counter.

(c) What are the problems of the above counter?

(8+4+3 marks) [CSE 205 | L-2/T-1 | 2012–13]

Q9. Draw and explain a four-bit binary count-up ripple counter with T flip-flops. **(12 marks)** [CSE 205 | L-2/T-1 | 2011–12]

Synchronous Counter Design

Q1. A ring counter is initialized incorrectly.

(a) Analyze the resulting sequence of states.

(b) Propose a circuit modification to ensure automatic correction.

(12 marks) [CSE 205 | L-2/T-1 | 2024–25]

Q2. Using JK flip-flops, design a synchronous counter that counts in the following sequence: $1, 2, 5, 7, 1, 2, \ldots$ Analyze the final circuit to ensure that if the circuit enters any of the unused states, after a finite number of clock cycles it comes to a used state. **(13 marks)** [CSE 205 | L-2/T-1 | 2022–23]

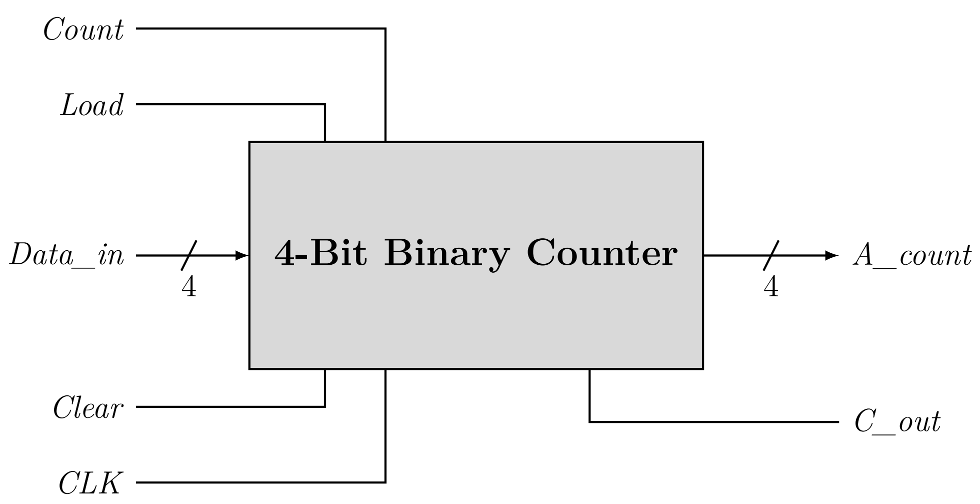
Q3. Design a synchronous counter that counts in the following sequence: $0 \to 2 \to 4 \to 7 \to 5 \to 3 \to 0 \to 2 \to 4 \to \ldots$ Design the counter using one JK flip-flop, one T flip-flop and one D flip-flop. Draw the circuit diagram. **(20 marks)** [CSE 205 | L-2/T-1 | 2021–22]

Q4. Design a synchronous counter that counts in the sequence $0 \to 2 \to 4 \to 7 \to 5 \to 3 \to 0 \to 2 \to 4 \to \cdots$ Design the counter using SR Flip-Flops and draw the circuit diagram. **(20 marks)** [CSE 205 | L-2/T-1 | 2020–21]

Q5. Design a synchronous counter that counts in the following sequence: $0 \to 1 \to 2 \to 5 \to 4 \to 6 \to 0 \to 1 \to 2 \to \cdots$ Design the counter using SR Flip-Flops and draw the circuit diagram. **(20 marks)** [CSE 205 | L-2/T-1 | 2017–18]

Q6. Design a synchronous counter with the sequence: $1 \to 4 \to 5 \to 7 \to 2 \to 1$. You need not take any measure for the unused states. **(10 marks)** [CSE 205 | L-2/T-1 | 2016–17]

Q7. Show two ways to design a Modulo-6 counter using a 4-bit binary counter and necessary basic gates. The Modulo-6 counter repeatedly goes through $0, 1, 2, 3, 4, 5$.



(12 marks) [CSE 205 | L-2/T-1 | 2015–16]

Q8. Design a 3-bit binary up counter that counts $1 \to 2 \to 5 \to 7 \to 1 \cdots$, where any invalid state goes to the immediate next valid state in the sequence. Use T Flip-Flops. Show the state diagram, state table, function equations, minimization (if necessary) and the circuit diagram. **(15 marks)** [CSE 205 | L-2/T-1 | 2014–15]

Q9. Design a 3-bit synchronous binary up counter that counts $1 \to 2 \to 4 \to 7 \to 1 \to \cdots$, where any invalid state goes to the immediate next state in the sequence. Use JK flip-flops. Show the state table, state diagram, function minimization (if necessary), and circuit diagram. **(20 marks)** [CSE 205 | L-2/T-1 | 2013–14]

Q10. Design a synchronous counter with the following binary sequence: $0, 1, 3, 7, 6, 4$ and repeat. Use T flip-flops. **(15 marks)** [CSE 205 | L-2/T-1 | 2012–13]

Synchronous Gray Code Counter

Q1. Design a 3-bit synchronous Gray Code Counter using JK Flip-Flops. Show the state diagram, state table, function equations and the circuit diagram. **(15 marks)** [CSE 205 | L-2/T-1 | 2014–15]

Q2. Develop a synchronous 3-bit counter with a Gray code sequence using JK flip-flops.
(Hint: Gray code count sequence for a 2-bit counter is: $00, 01, 11, 10, 00, 01, \ldots$*)* **(18 marks)** [CSE 205 | L-2/T-1 | 2011–12]

Switch-Tail Ring Counter

Q1. Design a 5-bit ring counter using a shift register and illustrate its operation using a timing diagram. **(10 marks)** [CSE 205 | L-2/T-1 | 2017–18]

Q2. Draw the logic diagram of a four-stage switch-tail ring counter. **(5 marks)** [CSE 205 | L-2/T-1 | 2011–12]

Johnson Counter vs Ring Counter

Q1. Write a short note about a ring counter. **(5 marks)** [CSE 205 | L-2/T-1 | 2022–23]

Q2. Distinguish between a ring counter and a Johnson counter. Design a 5-bit ring counter and a 5-bit Johnson counter using shift registers and illustrate their operation using timing diagrams. **(10 marks)** [CSE 205 | L-2/T-1 | 2020–21]

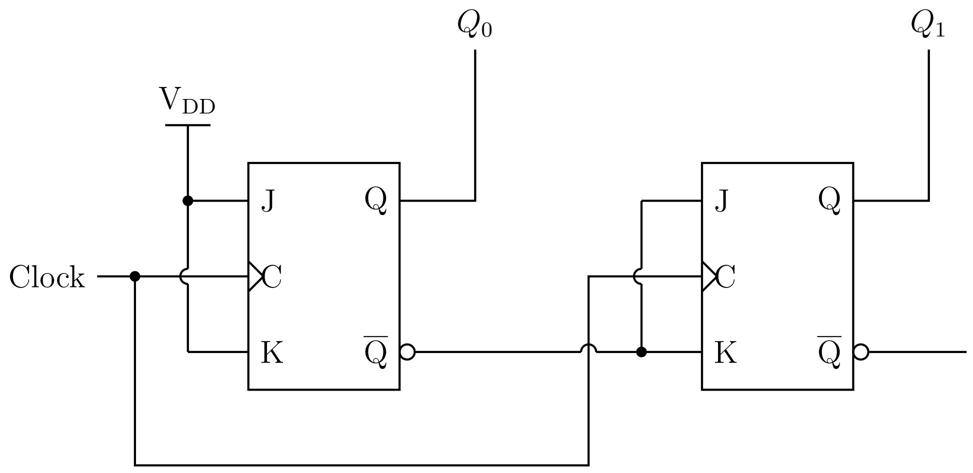
↪ *Similar: 2019–20*

Q3. Design a 4-bit Johnson counter using a shift register and illustrate its operation using a timing diagram. **(7 marks)** [CSE 205 | L-2/T-1 | 2018–19]

Q4. Design a 4-bit Johnson Counter. What is the disadvantage of a Johnson Counter? What is the difference between a Johnson Counter and a Ring Counter? **(10 marks)** [CSE 205 | L-2/T-1 | 2014–15]

Counter Direction Analysis

Q1. A counter circuit has V_{DD} assigned to Logic 1; the current values of LSB Q_0 and MSB Q_1 are 0 and 1 respectively. Find the direction (up or down) in which the circuit counts when the next clock pulse occurs. What would have to be altered to make the circuit count in the other direction?



(7 marks) [CSE 205 | L-2/T-1 | 2015–16]

Clock Generator / Divider

Q1. Given a 100 MHz clock signal, derive a circuit using D flip-flops to generate 50 MHz and 25 MHz clock signals. Draw a timing diagram for all three clock signals. **(10 marks)** [CSE 205 | L-2/T-1 | 2017–18]

Universal Shift Register

Q1. Design a 4-bit universal shift register using multiplexers and D flip-flops. **(13 marks)** [CSE 205 | L-2/T-1 | 2024–25]

Q2. Compare the hardware efficiency and flexibility of universal shift registers with dedicated shift registers in complex digital systems. **(7 marks)** [CSE 205 | L-2/T-1 | 2024–25]

Q3. Use a 2-to-4 decoder, NAND gates and edge triggered D flip-flops to design a 4-bit shift register module that has the following functions:

S_1	S_0	Mode
0	0	Shift right (all 4 bits)
0	1	Shift left (all 4 bits)
1	0	Synchronous common clear
1	1	Synchronous parallel load

(10 marks) [CSE 205 | L-2/T-1 | 2021–22]

Q4. Draw the logic diagram of a 3-bit universal shift register with mode selection inputs s_2, s_1, s_0. The register operates with eight modes: (000) No change, (001) Invert all bits, (010) Set all bits to 0, (011) Set all bits to 1, (100) Parallel load, (101) Shift left, (110) Shift right, (111) No change. **(15 marks)** [CSE 205 | L-2/T-1 | 2016–17]

Q5. Draw the logic diagram of a two-bit register with two D flip-flops and two 4×1 multiplexers with mode selection inputs s_1 and s_0. The register operates with four modes: (00) No change, (01) Complement the two outputs, (10) Clear register to 0 (synchronous), (11) Load parallel data. **(12 marks)** [CSE 205 | L-2/T-1 | 2015–16]

Q6. Draw the circuit diagram of a 4-bit universal shift register using JK flip-flops. **(12 marks)** [CSE 205 | L-2/T-1 | 2012–13]

Serial Adder & Serial Subtractor

Q1. A 4-bit shift register is used for serial data transmission.

(a) Analyze the timing relationship between input data, clock and output.

(b) Explain how errors may propagate in serial communication using shift registers.

(16 marks) [CSE 205 | L-2/T-1 | 2024–25]

Q2. Design a sequential circuit to copy the content of shift register A to shift register B. Use a block diagram to represent the shift register in the diagram. **(12 marks)** [CSE 205 | L-2/T-1 | 2022–23]

Q3. Design a serial subtractor that will perform the operation $A - B$, where $A = a_{n-1} \dots a_1 a_0$ and $B = b_{n-1} \dots b_1 b_0$. The operands are applied to the serial subtractor sequentially, beginning with bits a_0 and b_0. Use JK flip-flops in your design. **(10 marks)** [CSE 205 | L-2/T-1 | 2021–22]

Q4. Design a serial adder that computes the sum of two 4-bit binary numbers A and B stored in individual shift registers. The result of the addition is stored in the shift register representing A. **(10 marks)** **[CSE 205 | L-2/T-1 | 2017–18, 2016–17]**

Q5. Design a serial subtractor that subtracts the content of register B from the content of register A, where the contents of A and B are stored in individual shift registers. The result of the subtraction is stored in the shift register representing A. (Both A and B are 4-bit binary numbers.) **(10 marks)** **[CSE 205 | L-2/T-1 | 2013–14]**

Q6. Draw the logic diagram of a serial adder. Show the values of inputs and outputs of all components (e.g., flip-flop, register) used in the design after every clock pulse while adding the following two numbers: 0101 and 0011. **(6+6 marks)** **[CSE 205 | L-2/T-1 | 2011–12, 2012–13]**

BUET · CSE 205 · Digital Logic Design

Digital Logic Design

Memory & Programmable Logic

RAM, ROM, Address Multiplexing, PLA Design

S8 Memory & Programmable Logic

ROM Design & Expansion

Q1. You have several 32×4 ROMs in the laboratory. Each ROM can be enabled/disabled using a 1/0 signal respectively. How would you implement a 128×4 ROM using the available ROMs? You may use any additional circuitry if needed. **(10 marks)** **[CSE 205 | L-2/T-1 | 2013–14]**

PLA Design

Q1. Design a PLA to implement the following functions with the minimization of the number of product terms:

$$F_1(x, y, z) = \Sigma(1, 3, 5, 6), \qquad F_2(x, y, z) = \Sigma(0, 1, 6, 7)$$
$$F_3(x, y, z) = \Sigma(3, 5), \qquad F_4(x, y, z) = \Sigma(1, 2, 4, 5, 7)$$

(15 marks) **[CSE 205 | L-2/T-1 | 2024–25]**

Q2. Tabulate the PLA programming table for the two Boolean functions listed below. Minimize the numbers of product terms:

$$F_1(A, B, C) = \Sigma(0, 1, 2, 4)$$
$$F_2(A, B, C) = \Sigma(0, 5, 6, 7)$$

(15 marks) **[CSE 205 | L-2/T-1 | 2022–23]**

Q3. A combinational circuit is defined by the following two functions:

$$F_1(A, B, C) = \Sigma(3, 5, 6, 7)$$
$$F_2(A, B, C) = \Sigma(0, 2, 4, 7)$$

The circuit needs to be implemented with a PLA having three inputs, four product terms, and two outputs. Show the PLA programming table. **(15 marks)** **[CSE 205 | L-2/T-1 | 2021–22]**

Q4. Design a digital security lock using PLA which takes BCD as input and opens after converting the input to Excess-3 code. Use the minimum number of product terms. (Show the fuse map.) **(15 marks)** **[CSE 205 | L-2/T-1 | 2020–21]**

Q5. Design a circuit for the following functions using a Programmable Logic Array (PLA):

$$A(x, y, z) = \Sigma(0, 1, 2, 5, 7)$$
$$B(x, y, z) = \Sigma(0, 1, 3, 6, 7)$$

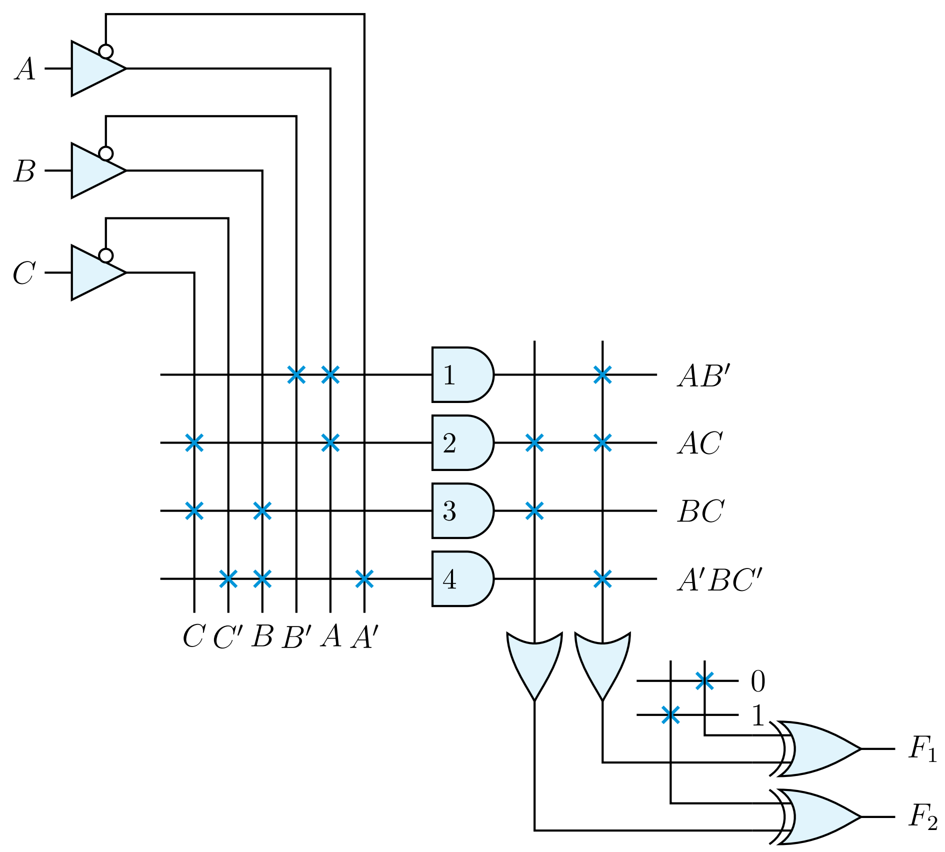
(10 marks) **[CSE 205 | L-2/T-1 | 2019–20]**

Q6. Design a security lock using PLA which will be opened using the following combinations:

$$A(x, y, z) = \Sigma(1, 3, 5, 6), \quad B(x, y, z) = \Sigma(0, 1, 6, 7)$$
$$C(x, y, z) = \Sigma(3, 5), \quad D(x, y, z) = \Sigma(1, 3, 4, 5, 7)$$

(12 marks) **[CSE 205 | L-2/T-1 | 2018–19]**

Q7. For a given Programmable Logic Array (PLA), find the function expressions for F_1 and F_2 and show the PLA programming table.



(13 marks) [CSE 205 | L-2/T-1 | 2015–16]

Q8. Write the difference between ROM and PLA. A combinational circuit is defined by the following functions:

$$F_1(A, B, C) = \Sigma\,(0, 1, 2, 4)$$
$$F_2(A, B, C) = \Sigma\,(0, 5, 6, 7)$$

Implement the circuit with a PLA having three inputs, four product terms and two outputs. **(5+10 marks)** [CSE 205 | L-2/T-1 | 2014–15]

Q9. Derive the PLA program table for a combinational circuit that squares a 3-bit binary number. Minimize the number of product terms. **(25 marks)** [CSE 205 | L-2/T-1 | 2013–14]

Q10. Implement the following three Boolean functions with a PLA:

$$F_1 = \Sigma\,(0, 1, 2, 4)$$
$$F_2 = \Sigma\,(0, 5, 6, 7)$$
$$F_3 = \Sigma\,(0, 3, 5, 7)$$

(15 marks) [CSE 205 | L-2/T-1 | 2011–12]

RAM Design

Q1. If you use two dimensional decoding for addressing a 1K word memory, can you optimize the cost? Justify your answer. **(8 marks)** [CSE 205 | L-2/T-1 | 2024–25]

Synchronous Sequential Circuit Design

Q1. Design an automatic vending machine, when the candy bars inside the machine cost 3 Tk and the machine accepts 1 Tk and 2 Tk only. The system accepts the bills sequentially. The circuit produces a level output that delivers the candy whenever the amount received by the machine is 3 Tk or more. The excess amount is kept deposited for the next candy. Design a synchronous sequential circuit for the vending machine using S-R flip-flops. Provide the circuit diagram as well. **(25 marks)** [CSE 205 | L-2/T-1 | 2024–25]

Q2. Design a simplified traffic-light controller that switches traffic lights on a crossing where a north-south (NS) street intersects an east-west (EW) street. The input to the controller is the WALK button pushed by pedestrians who want to cross the street. The outputs are two signals NS and EW that control the traffic lights in NS and EW directions. When NS or EW are 0, the red light is on and when they are 1, the green light is on. When there are no pedestrians, NS = 0 and EW = 1 for 1 minute, followed by NS = 1 and EW = 0 for 1 minute and so on. When a WALK button is pushed, NS and EW both come 1 for a minute when the present minute expires. After that the NS and EW signals continue alternating. For the traffic-light controller: (i) Develop a state diagram and state/output table. (ii) Find the transition and excitation tables using S-R flipflops. (iii) Draw a circuit diagram. **(23 marks)** [CSE 205 | L-2/T-1 | 2023–24]

Q3. Design a divide-by-three counter which counts the number of '1's from an input sequence of binary stream. This counter outputs one 1 for every three 1's seen as input (not necessarily in succession). After outputting a 1, it starts counting all over again. Design the counter using D flip-flops. **(12 marks)** [CSE 205 | L-2/T-1 | 2023–24]